DOOMSCROLLING DETECTION AND DIGITAL MINDFULNESS MOBILE APPLICATION FOR SHORT-FORM VIDEO PLATFORMS USING VADER AND FUZZY LOGIC

by

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LIST OF ABBREVIATIONS

API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
CoG	Center of Gravity
DSI	Doomscroll Severity Index
IPO	Input–Process–Output
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
JITAI	Just-in-Time Adaptive Intervention
MVL	Minimum Viable Lexicon
NSD	Negative Sentiment Density
OOV	Out-of-Vocabulary
PEOU	Perceived Ease of Use
PU	Perceived Usefulness
RAM	Random Access Memory
SME	Subject Matter Expert
SUS	System Usability Scale
TAM	Technology Acceptance Model
UI	User Interface
VADER	Valence Aware Dictionary and sEntiment Reasoner
VLM	Vision-Language Model
VLOPs	Very Large Online Platforms
VQA	Visual Question Answering

ABSTRACT

Doomscrolling on short-form video platforms is difficult to address with time-based digital well-being tools alone because risk may depend on both prolonged engagement and exposure to negative content. This study designed, developed, and evaluated a privacy-preserving Android application for heuristic doomscrolling-related risk estimation on TikTok, Facebook Reels, and Instagram Reels. The system uses Android Accessibility Service monitoring, local session tracking, VADER-compatible sentiment analysis with a Filipino/Taglish Minimum Viable Lexicon, a Moondream 0.5B no-text visual fallback, and a fuzzy inference engine to produce session risk scores and week-level Doomscroll Severity Index values. A two-week pilot field evaluation was conducted with 50 adult Filipino Android users assigned evenly to intervention and logging-only control groups. Results showed 10,134 logged sessions, with 87.0% sentiment-reliable coverage. Between-group Week 1-to-Week 2 comparisons favored the intervention group for session duration, video dwell time, Negative Sentiment Density, and Doomscrolling Scale scores. Week 1 DSI showed strong baseline convergence with self-reported doomscrolling, although session duration alone had the highest rank association. User evaluation met all ISO/IEC 25010, SUS, and TAM targets, and both subject matter experts rated the system favorably. The study concludes that the system is a feasible non-clinical prototype for privacy-preserving short-form-video self-monitoring, while further validation and longer deployment are needed.

Keywords: doomscrolling, digital mindfulness, fuzzy logic, VADER, sentiment analysis, Android, short-form video, privacy-preserving computing

Chapter 1

INTRODUCTION

Doomscrolling refers to the compulsive consumption of distressing or negative online content despite its adverse emotional and behavioral effects (Sharma et al., 2022). Mainstream device- and platform-level digital well-being tools still emphasize screen-time summaries, app timers, or break reminders instead of session-level interpretation of fast-changing feed content (Apple Inc., 2025; Google, 2024; Mosseri, 2021; Rahmillah et al., 2023; TikTok, 2024). This thesis addresses that gap through the design, development, and evaluation of the **Heuristic Risk-State Estimation System** — a privacy-preserving Android app that performs on-device risk estimation using behavioral signals, a text-first sentiment path, and an on-device Vision-Language Model (VLM) fallback for no-text items. Here, "detection" refers to computational risk estimation from observable proxies, not diagnostic classification.

The system combines threshold-based heuristics, interpretable fuzzy rule-based inference (Pickering et al., 2025; Vashishtha et al., 2023), a VADER-compatible text-first sentiment pipeline for low-resource code-mixed text (Hutto & Gilbert, 2014; Mohammed & Prasad, 2023; Nazir et al., 2026), and an on-device no-text VLM fallback implemented with Moondream 0.5B (Das & Singh, 2023; Sharshar et al., 2025). All processing occurs locally on the user's device to preserve privacy under real-world mobile constraints. It is evaluated as a bounded software-engineering study covering software quality, user acceptance, expert review, baseline convergent association, and short-term behavioral differences during a two-week field deployment.

1.1. Background of the Study

In recent years, doomscrolling has emerged as a behavioral concern in algorithm-driven social media environments. Short-form platforms such as TikTok, Facebook Reels, and Instagram Reels are built around rapid, recommendation-driven streams of content, and prior studies associate heavier or more compulsive use with attention-related strain, procrastination, reduced mindfulness, anxiety-related symptoms, and related negative outcomes (Canila et al., 2023; Cardoso et al., 2024; Hawwa et al., 2025; Qin et al., 2022; Taskin et al., 2024; Zhang, 2023).

Existing digital well-being tools address this problem only partly. Prior work shows that the emotional effects of scrolling depend on what users are exposed to, not just how long they scroll (Buchanan et al., 2021). Mainstream responses still center on time limits, usage summaries, or fixed break reminders, while app-level interventions reviewed in the literature remain heterogeneous and do not foreground privacy-by-design as a core requirement (Apple Inc., 2025; Google, 2024; Mosseri, 2021; Rahmillah et al., 2023; Tewari, 2023; TikTok, 2024). This suggests a narrower engineering gap for the present study: whether a private mobile tool can move beyond time alone by combining observable interaction patterns with a text-first sentiment proxy while still handling no-text items through an on-device visual fallback.

Mindfulness-oriented digital interventions provide limited support for brief app-delivered self-regulation prompts (Mitsea et al., 2023). Just-in-time smartphone interventions have also shown that smartphone-based support can be delivered in the moment, with feasibility and acceptability depending on timing, receptivity, and burden management (Mair et al., 2022; Roffarello & De Russis, 2021; Teepe et al., 2021; Yang et al., 2023). These works support in-the-moment prompting; what remains untested is a single system that continuously monitors short-form video behavior, processes extracted text locally, resolves no-text items through a visual fallback, and issues adaptive prompts entirely on-device.

In the Philippine context, local peer-reviewed studies indicate that doomscrolling, problematic social-media use, and digital well-being are relevant concerns, although the strongest available evidence remains student-focused (Canila et al., 2023; Cardoso et al., 2024; Punzalan et al., 2024). Nearby adult Philippine contexts are more indirect, including social-media use among older adults and anxiety-related social-media exposure findings in adult workers (Castillo et al., 2022; Zamora et al., 2021). The local literature therefore supports contextual relevance, but not adult prevalence, validated thresholds, or expected intervention effects. Against that backdrop, the study examines whether the system can combine behavioral monitoring, a text-first sentiment proxy, the no-text VLM fallback, and adaptive prompting within a privacy-by-design architecture, then be evaluated for usability, reliability, and practical usefulness among adult Filipino users.

1.2. Statement of the Problem

This study addresses the gap left by time-based or platform-controlled digital well-being tools by asking whether a privacy-preserving Android app can be designed, developed, and evaluated for doomscrolling-related risk estimation on short-form video platforms. Current responses remain fragmented: platform-native tools are usually time-based, while related research analogues rarely combine local monitoring, content-sensitive estimation, and adaptive prompting in one deployable adult short-form video system. Because the available local literature does not yet support calibrated estimator thresholds or clinical interpretation, the problem is framed as a software-engineering question. Specifically, it seeks to answer the following questions:

1. What privacy-preserving mobile architecture and estimation framework can support doomscrolling-related risk estimation on the target short-form video

platforms using behavioral indicators and sentiment-related indicators when reliably resolvable, with 2-input behavioral fallback for sentiment-unreliable sessions?

2. What short-term Week 1-to-Week 2 changes are observed in selected logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, and within the intervention group across the same period?
3. What baseline convergent association exists between the fixed-prior Week 1 Doomscroll Severity Index (DSI) and participants' self-reported doomscrolling scores among participants with at least three sentiment-reliable Week 1 sessions?
4. How do users evaluate the system using the ISO/IEC 25010 software quality model and the Technology Acceptance Model (TAM)?
5. How do subject matter experts evaluate the system's technical design, privacy safeguards, heuristic logic, and intervention structure?

1.3. Significance of the Study

Contribution to the Body of Knowledge: This study contributes a mobile-systems design and pilot-scale field evaluation of one app that integrates behavioral monitoring, a text-first sentiment proxy, a no-text VLM fallback, adaptive prompting, and on-device privacy. Its contribution is an engineering evaluation framework for feasibility, acceptability, and estimator plausibility. The **Minimum Viable Lexicon (MVL)** workflow supports low-resource, code-mixed text handling within the system.

Beyond its academic contribution, the study may offer value to the following stakeholders:

Adult users of short-form video platforms may benefit from a consent-based self-monitoring application that supports reflective interruption of problematic scrolling behavior without sending raw content to external servers.

Software developers and mobile-computing researchers may reuse the app's architecture, hybrid text/VLM proxy design, and fallback logic when building privacy-preserving digital well-being systems for code-mixed social media environments.

Digital well-being researchers may use the study as an example of how behavioral logging, lightweight sentiment analysis, and adaptive prompting can be combined in a mobile field evaluation.

Educational institutions and digital well-being advocates may use the completed study output to support responsible discussions about privacy-preserving, user-directed self-regulation tools.

1.4. Objectives of the Study

General Objective:

To design, develop, and evaluate the proposed system as a privacy-preserving, on-device Android application for doomscrolling-related risk estimation on short-form video platforms.

Specific Objectives:

1. To design and develop the system to estimate doomscrolling-related risk using behavioral indicators and sentiment-related indicators when reliably resolvable, with 2-input behavioral fallback for sentiment-unreliable sessions.
2. To determine the short-term Week 1-to-Week 2 changes in selected logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, and within the intervention group across the same period.

3. To determine the baseline convergent association between the fixed-prior Week 1 Doomscroll Severity Index (DSI) and participants' self-reported doomscrolling scores among participants with at least three sentiment-reliable Week 1 sessions.
4. To evaluate the computational efficiency of the system's core runtime algorithms through analytic time-complexity analysis of VADER scoring, per-item routing, VLM fallback, fuzzy inference, NSD aggregation, week-level DSI computation, and quantile personalization, and to evaluate the system from the users' perspective using the ISO/IEC 25010 software quality model and TAM.
5. To obtain subject matter expert evaluation of the system's technical design, privacy safeguards, heuristic logic, and intervention structure.

1.5. Scope, Delimitations, and Limitations

This study covers the design, development, and evaluation of the system. The scope centers on the software artifact, its risk-estimation variables, and the two-week field evaluation.

Scope:

- Development of a native Android application with Android 8.0 (API level 26) as the minimum supported operating-system version. The screenshot-assisted no-text VLM path requires Android 11 (API level 30) or higher; devices below this version use the sentiment-unreliable, 2-input behavioral fallback when screenshot capture is unavailable.
- Implementation and evaluation cover one deployable system and a pilot-scale two-week field study involving 50 adult Filipino Android users assigned equally to an intervention group and a logging-only control group. Both groups completed a one-week baseline phase with prompts disabled. During the second week,

adaptive prompts were enabled only for the intervention group, while the control group continued logging without prompts. Aggregate usage and sentiment-related metrics were recorded throughout both weeks, followed by post-usage user evaluation.

- Use of the Android Accessibility Service API for text extraction, interaction-event monitoring, and screenshot-assisted no-text routing through `AccessibilityService.takeScreenshot`, with related interaction signals used to estimate session duration, compute video dwell time from content transitions, and process sentiment-related indicators from visible text or no-text visual items. Swipe counts are retained only as supporting logs.
- Design of a modular architecture with separation between data collection, processing, and user interface components.
- Implementation of the core estimation framework using threshold-based rules, text-first VADER sentiment analysis, an on-device no-text VLM fallback using Moondream 0.5B, and fuzzy-logic inference, with unresolved cases handled through sentiment-unreliable classification and 2-input behavioral fallback.
- The system is designed for TikTok, Facebook Reels, and Instagram Reels as target platforms, although actual platform exposure during the field deployment varied by participant and empirical claims are limited to platforms that yielded stable extraction.
- On-device processing, where raw text and temporary no-text screen frames are processed locally in RAM and discarded after scoring, while only aggregate local metrics and configuration data are retained on-device.
- A user-directed design in which risk estimates and prompts are delivered only to the consenting user on their own device, with no remote administrator dashboard.

- Evaluation uses the ISO/IEC 25010 Software Quality Model (Functional Suitability, Performance Efficiency, Usability, Reliability), the Technology Acceptance Model (Perceived Usefulness, Perceived Ease of Use), subject matter expert review, and short-term observed differences under the study conditions.

Delimitations and Limitations:

This study evaluates software behavior and short-term usage differences, not clinical outcomes. The following boundaries include both deliberate choices made to define the study's coverage and constraints that may affect the findings or their generalizability. Changes in logged metrics should not be interpreted as reductions in anxiety, distress, or mental health conditions.

- The study targets Filipino Android users aged 18 years and above who actively use at least one of the target short-form video platforms. Its external validity does not extend to minors, iOS users, or users who primarily consume long-form or non-short-form content.
- Any field deployment is limited to Android devices that can install the study build, maintain the required permissions, and keep the monitoring service sufficiently stable during the evaluation window.
- The risk-estimation framework addresses only short-form video platforms and does not cover long-form video, news websites, general web browsing, or other social media formats outside TikTok, Facebook Reels, and Instagram Reels.
- The study is not designed as a clinical diagnostic system, medical intervention, psychotherapy tool, or long-term habit-formation trial.
- The study does not claim to prevent doomscrolling. It evaluates a heuristic risk-estimation system and observes short-term behavioral differences between intervention and control groups.

- The study does not attempt multi-site rollout, large-sample validation, or extensive model optimization.
- The study excludes covert monitoring, employer surveillance, parental-control workflows, remote dashboards, and third-party alerting mechanisms.
- The use of Android Accessibility Service to read on-screen content from third-party applications may create platform-policy concerns. Ethical risk is reduced through explicit user consent, local processing, and the absence of third-party surveillance workflows, but policy changes remain outside the researchers' control.
- The Accessibility Service approach is fragile by nature. Changes in the user interface of TikTok, Facebook, or Instagram may disrupt extraction logic, and Android device manufacturers may aggressively terminate background services, affecting data continuity.
- Continuous monitoring may affect battery life, thermal behavior, and background stability on some devices, which can in turn influence both system reliability and user acceptance.
- The system does not perform continuous full-resolution video or audio analysis. It relies on behavioral logs, accessibility-extracted text, and sparse transient on-device screen-frame sampling for no-text items. The system therefore still estimates risk from observable proxies rather than from full multimodal understanding of the viewed media.
- Although the deployable app supports Android 8.0 (API 26) and higher, the screenshot-assisted no-text VLM path depends on `AccessibilityService.takeScreenshot`, which is available only on Android 11 (API 30) and higher. On older compatible devices, no-text visual resolution

degrades to the sentiment-unreliable / 2-input fallback path when screenshot capture is unavailable.

- Video duration is not consistently exposed across all target applications through the Accessibility Service. For this reason, the system relies on video dwell time and related interaction patterns rather than exact platform-reported video length.
- Because Android Accessibility APIs do not expose a guaranteed content-level identifier for third-party short-form feed items, transition detection remains heuristic and may still mis-segment some consecutive items, especially when they are text-barren or visually similar.
- The baseline sentiment engine is English-centric. Although the system extends it with a limited Filipino/Taglish lexicon, unsupported dialects, slang drift, and code-mixed text may still reduce sentiment reliability.
- Sessions with high out-of-vocabulary text, or no-text items that still cannot be resolved reliably after VLM routing, are treated as sentiment-unreliable and excluded from some sentiment-dependent analyses, which may reduce the effective analyzable sample for some outcomes.
- Self-reported profile data and Doomscrolling Scale responses remain subject to recall bias and social desirability bias.
- Because recruitment is limited to consenting participants willing to install a monitoring application, the sample may overrepresent users who are already receptive to self-regulation tools.
- Although the logging-only control group was intended to isolate prompt effects from monitoring awareness and natural fluctuation, the study remains non-blinded: intervention-group participants know they are receiving prompts, which may introduce expectancy or novelty effects beyond the prompt mechanism itself.

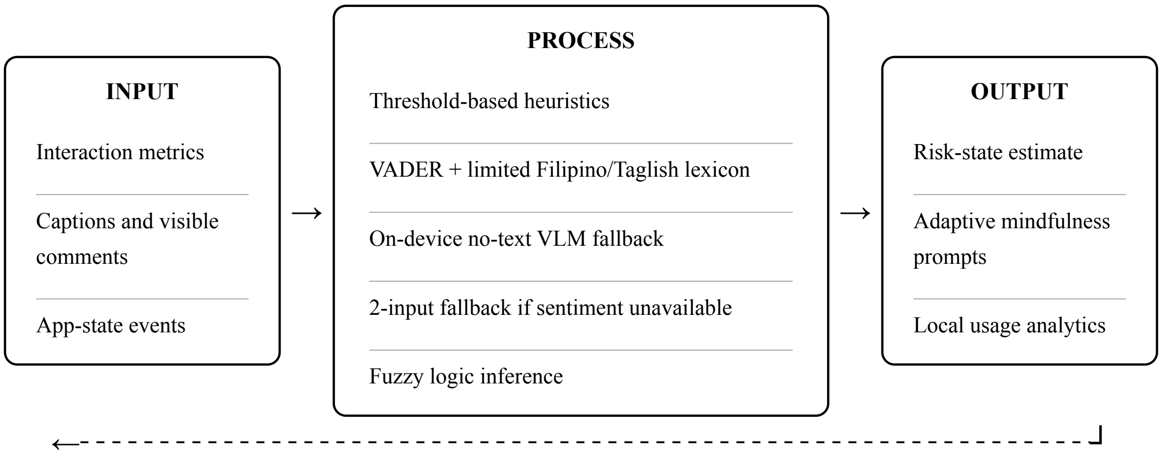
- For the intervention group, prompt-excluded active-use metrics can be mechanically lower by design when prompts occur because prompt-display time is removed from session duration and dwell-time calculations. For this reason, Chapter 3 treats raw elapsed session duration and raw elapsed dwell time as the primary behavioral comparison, while prompt-excluded metrics are kept only as supplementary traces of prompt-interrupted use.
- The two-week deployment window captures only short-term behavior. It is insufficient for demonstrating long-term habit formation, retention, or sustained behavioral change.
- Participants know they are being monitored, which introduces Hawthorne-effect risk and may reduce the naturalism of observed behavior.

1.6. Conceptual Framework

The study uses the **Input-Process-Output (IPO) model** to illustrate the implemented runtime architecture and data flow for heuristic risk estimation (Laudon & Laudon, 2022). The IPO model was selected because it provides a compact and transparent representation of how observable behavioral and content signals are transformed into a heuristic risk estimate and, in the intervention phase, into an adaptive prompt. The three stages of the model are used in this study to describe (a) the runtime signals that the mobile application collects on-device, (b) the reasoning pipeline that combines behavioral thresholds, sentiment analysis, and fuzzy logic inference to derive a graded risk state, and (c) the outputs that either inform local analytics only (Week 1 baseline) or additionally trigger an adaptive mindfulness prompt (Week 2 intervention). The addition of a **feedback loop** in Figure 1 makes explicit that the pipeline is not one-shot: outputs produced during an ongoing scrolling session are fed back into the next interval's Input, allowing the system to continuously re-estimate risk based on subsequent user behavior after a prompt has been shown or after further scrolling has occurred.

Figure 1

Conceptual Framework of the Study Using the IPO Model with Feedback Loop



Feedback loop: prompt response and continued behavior feed the next interval's Input for re-estimation

The IPO model here refers to the runtime application pipeline rather than the full research workflow. The **Input** stage captures three families of signals directly on-device: (i) interaction metrics such as swipe or scroll transitions, video dwell time, and session-level engagement counters; (ii) captions and visible comments extracted from the target short-form video platforms; and (iii) app-state events that mark session start, foreground and background transitions, and prompt-response events. These signals are all collected locally through the Android Accessibility Service and never leave the device.

The **Process** stage combines four reasoning layers. Threshold-based heuristics screen the interaction metrics and produce membership degrees for Session Duration and Video Dwell Time. VADER, extended with a limited Filipino/Taglish minimum viable lexicon, computes item-level sentiment on usable text units. When a viewed item has no usable caption or visible comments, an on-device Vision-Language Model fallback (Moondream 0.5B) is invoked to estimate whether the item contributes negative exposure. If neither text nor VLM paths can resolve an item reliably, the pipeline degrades gracefully to a two-input fallback that relies only on Session Duration and Video Dwell Time. All resolved signals are then combined by a fuzzy logic inference engine that yields a graded RiskScore across low, warning, and critical bands.

The **Output** stage produces the RiskScore, the corresponding risk state, and — during Week 2 — an adaptive mindfulness prompt at level L1 (awareness), L2 (pause), or L3 (pause-and-reset short breathing break). Local usage analytics are also written to on-device storage for later export. The **feedback loop** shown in Figure 1 operates once the live duration gate is met and the live RiskScore enters the Warning or Critical bands: the app selects an intervention level, closes the current interval, and begins a new one only if the user returns to the target platform. The response to the prompt (dismissal, engagement, or platform exit) and the subsequent behavior become part of the next interval's Input, so the system continuously updates its risk estimate rather than relying on a single decision point. When sufficient reliable baseline sessions exist, the live prompt engine may personalize Session Duration and NSD memberships from Week 1 quantiles while Video Dwell Time remains fixed, further reinforcing the adaptive character of the loop.

1.7. Theoretical Framework

The theoretical framework of this study integrates four complementary theories that jointly justify each aspect of the system: the **Doomscrolling Feedback Loop Model** (Sharma et al., 2022), which anchors the phenomenon being estimated and grounds the specific role of Negative Sentiment Density; **Uses and Gratifications Theory** (Katz et al., 1973; Ruggiero, 2000), which explains why users initiate and prolong exposure to distressing short-form content; **Self-Determination Theory** (Ryan & Deci, 2000), which grounds the design of non-coercive, autonomy-supportive mindfulness prompts; and the **Dual-Systems / Habit Loop** perspective (Kahneman, 2011; Wood & Neal, 2007), which explains the automatic, cue-driven character of prolonged scrolling and the timing at which interruption is expected to be most effective. The evaluation of the resulting artifact is then guided by **ISO/IEC 25010** (International Organization for Standardization, 2023) and the **Technology Acceptance Model (TAM)** (Adnan et al., 2025), while the baseline-to-intervention design supports short-term behavioral comparison.

1.7.1. Doomscrolling Feedback Loop Model

The system is primarily informed by the **Doomscrolling Feedback Loop Model** derived from Sharma et al. (2022). The model conceptualizes doomscrolling as a self-reinforcing three-phase cycle:

1. **Antecedents or triggers:** Users are drawn toward distressing or uncertainty-inducing content by factors such as anxiety, fear of missing out, perceived information needs during crises, or algorithmic amplification of emotionally charged material. The salient feature of this phase is not merely that negative content is available, but that it is **densely present** within the user's immediate feed, which increases the probability that any given item consumed carries negative valence.
2. **Behavior:** Users engage in persistent, repetitive scrolling through negative or emotionally charged content. In the short-form video context this manifests as prolonged single-session engagement (long session duration) and reduced per-item dwell time as users rapidly sample many items, or, alternately, elevated per-item dwell time on distressing items.
3. **Outcome:** Sustained exposure to such content contributes to distress, negative affect, and heightened vigilance, which may in turn reinforce the antecedent phase (elevated anxiety) and the behavior phase (further compensatory scrolling), closing the loop.

The distinctive contribution of Sharma et al.'s model over generic screen-time or problematic-use accounts is precisely the loop structure: negative exposure is not treated as an incidental side-effect but as an active driver of the next iteration. This study operationalizes each phase with observable proxies. The behavior phase is estimated by **Video Dwell Time** and **Session Duration**, which are extracted from swipe/scroll transition signals and app-state events. The antecedent phase — the density of negative content in the user's immediate exposure — is estimated by **Negative Sentiment Density (NSD)**, which serves as the model's operational bridge into content-level measurement.

Role of Negative Sentiment Density (NSD) in the model. NSD is defined in this study as the proportion of resolvable content units within a session that are classified as negative, computed from analyzable caption and visible-comment text units that pass the text-side reliability screen, together with no-text items resolved through the on-device VLM fallback. NSD occupies the **antecedents/triggers** branch of the Doomscrolling Feedback Loop Model: it estimates how negatively-loaded the user's immediate content environment is, and therefore how strongly the environment is likely to trigger continued scrolling in the next interval. In the loop's forward direction, a rising NSD combined with elevated Session Duration and altered Video Dwell Time signals that the user is transitioning from ordinary engagement into a pattern more consistent with doomscrolling; the system responds with an adaptive prompt intended to interrupt the loop before the outcome phase is further reinforced. In the loop's return direction (Figure 1), the user's response to the prompt and the sentiment composition of items viewed after the prompt feed the next interval's NSD, so that NSD itself is continuously updated across the session rather than being computed once. NSD is therefore the theoretical framework's direct point of contact with the sentiment analysis subsystem: without an operational density measure at the antecedents phase, the loop model could only be observed at the behavior phase, and the interruption logic would collapse into a pure duration timer.

1.7.2. Uses and Gratifications Theory

Uses and Gratifications Theory (U&G) (Katz et al., 1973; Ruggiero, 2000) posits that media users actively select and continue engaging with content that satisfies specific psychological and social needs — information-seeking, mood management, surveillance, and social integration — rather than being passive recipients. In the context of short-form video platforms and doomscrolling, U&G explains **why** users initiate and prolong exposure to negative or distressing content: distressing content can paradoxically gratify surveillance and uncertainty-reduction needs during crises, and rapid item turnover on short-form platforms gratifies mood-repair and stimulation-seeking needs even when individual items are aversive. U&G therefore justifies treating scrolling behavior as goal-directed and non-random, which supports the study's use of behavioral proxies (Session Duration, Video Dwell Time) as meaningful indicators of engagement intensity rather than as noise. It also justifies designing prompts that offer an alternative gratification pathway — reflective interruption and self-monitoring — rather than merely blocking access, which would leave the underlying need unaddressed.

1.7.3. Self-Determination Theory

Self-Determination Theory (SDT) (Ryan & Deci, 2000) distinguishes between autonomous and controlled forms of motivation and identifies autonomy, competence, and relatedness as the three basic psychological needs whose support facilitates self-regulation and well-being. SDT grounds the design of the mindfulness prompts along two dimensions. First, the prompts are **non-coercive**: they present awareness, pause, or short breathing-break invitations at graded levels (L1, L2, L3) rather than forcing app closure, preserving user autonomy in deciding whether to continue. Second, the prompts are **informational rather than controlling**: they surface behavioral signals (elapsed session duration, current risk band) so that users can exercise competence in interpreting and acting on their own usage, which is consistent with SDT's finding that autonomy-supportive interventions are more likely to be internalized and sustained than externally imposed restrictions. SDT thereby justifies the study's choice of adaptive mindfulness prompts over hard blocks or punitive time limits.

1.7.4. Dual-Systems Reasoning and the Habit Loop

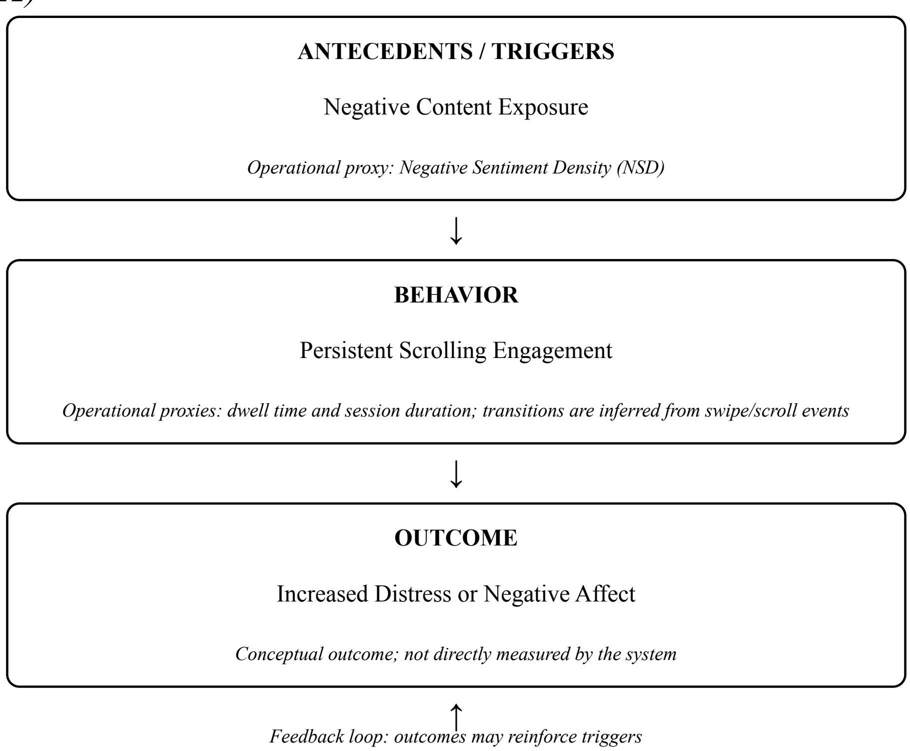
The **Dual-Systems** perspective (Kahneman, 2011) distinguishes between System 1 — fast, automatic, cue-driven processing — and System 2 — slow, effortful, reflective processing. The **Habit Loop** view of habitual behavior (Wood & Neal, 2007) complements this by describing habits as cue-routine-reward sequences in which contextual cues automatically activate learned motor and attentional routines with minimal deliberative control. Prolonged scrolling on short-form video platforms exemplifies System-1, cue-driven behavior: opening the app functions as the cue, the swipe-and-consume sequence is the routine, and brief affective spikes from novel content function as intermittent reward. This perspective grounds two design decisions in the present study. First, it justifies **timing** interruption around the behavioral signals themselves (a duration gate combined with the fuzzy risk band) rather than at fixed clock times, because a well-timed cue must arrive while the automatic routine is active in order to shift processing toward System 2. Second, it explains **why brief reflective prompts are expected to be effective at all**: by inserting a small deliberative moment into the loop, the prompt aims to convert an otherwise System-1 continuation decision into a System-2 evaluation, which is precisely the mechanism the mindfulness literature invokes for attentional reset.

1.7.5. Integration and Evaluation

Taken together, the four theories map onto the framework as follows: the Doomscrolling Feedback Loop Model defines **what** the system estimates and how NSD, Session Duration, and Video Dwell Time correspond to its phases; Uses and Gratifications explains **why** users engage with the behavior in the first place and why behavioral proxies are meaningful; Self-Determination Theory grounds the **form** of the intervention as autonomy-supportive rather than restrictive; and the Dual-Systems/Habit-Loop perspective grounds the **timing and mechanism** by which a brief prompt is expected to interrupt automatic scrolling. The framework thus justifies using observable proxies and non-clinical prompts. The evaluation of the resulting artifact is then guided by ISO/IEC 25010 (International Organization for Standardization, 2023) and TAM (Adnan et al., 2025), while the baseline-to-intervention design supports short-term behavioral comparison.

Figure 2

Doomscrolling Feedback Loop Diagrammatic Interception Model, adapted from Sharma et al. (2022)



The system is therefore a **heuristic risk-estimation tool** that uses behavioral and exposure proxies rather than direct measures of internal psychological states.

1.8. Definition of Terms

Adaptive Digital Wellness Prompts - Non-clinical system-generated prompts, such as awareness notifications, pause prompts, or short pause-and-reset breathing breaks, that are triggered according to estimated activity pattern severity to encourage reflective interruption of scrolling behavior.

Digital Mindfulness - A practice of intentional and self-aware technology use that promotes attention, presence, and self-regulation in digital environments (Aggarwal et al., 2024).

Doomscrolling - The compulsive and continuous consumption of negative or distressing content on digital platforms, particularly in ways that may reinforce anxiety, vigilance, or maladaptive engagement (Sharma et al., 2022).

Doomscrolling-Related Risk Estimation - The computational estimation of elevated scrolling risk using observable behavioral and sentiment-related proxies rather than direct measurement of internal psychological states.

Edge Computing - A computing approach in which data processing occurs on or near the data source - in this study, on the user's mobile device - rather than in remote cloud infrastructure, thereby reducing latency and privacy risks (Swathi et al., 2025).

Fuzzy Logic System - A computational approach that uses partial membership and rule-based reasoning to model ambiguous behavioral patterns, allowing graded risk estimation instead of binary classification through interpretable linguistic rules (Pickering et al., 2025; Vashishtha et al., 2023).

Heuristic Risk-State Estimation System - The Android application developed in this study for heuristic doomscrolling-related risk estimation and adaptive prompting.

ISO/IEC 25010 - An international software quality model used in this study to evaluate selected quality characteristics of the system, specifically Functional Suitability, Performance Efficiency, Usability, and Reliability (International Organization for Standardization, 2023).

Minimum Viable Lexicon (MVL) - The limited Filipino/Taglish lexicon extension used by the system to supplement VADER for recurring code-mixed social-media terms. It includes Filipino review-instrument candidate terms with implementation runtime valence values and separate neutral Filipino function-word entries used only to reduce false OOV inflation.

Negative Sentiment Density (NSD) - The proportion of resolvable content units within a session that are classified as negative. In this study, it is computed from analyzable caption and visible-comment text units that pass the text-side reliability screen together with no-text items resolved through the VLM fallback. It is used as a limited proxy for negative exposure.

Out-of-Vocabulary (OOV) Ratio - The proportion of extracted text tokens that are not recognized by the base VADER lexicon, emoji/emoticon handling, Filipino MVL extension, booster/negation rules, or neutral OOV-reduction list. High OOV values are used as a reliability screen rather than as a sentiment score.

Technology Acceptance Model (TAM) - A framework used to assess user acceptance of digital systems, particularly through the constructs of Perceived Usefulness and Perceived Ease of Use, and still commonly applied in mHealth acceptability evaluation (Adnan et al., 2025).

VADER Sentiment Analysis - Valence Aware Dictionary and sEntiment Reasoner, a lexicon- and rule-based sentiment analysis method for social media text that is adapted in this study for limited code-mixed use (Hutto & Gilbert, 2014).

Vision-Language Model (VLM) Fallback - A limited on-device visual-analysis path used in this study when a viewed short-form item has no usable caption or visible comments. It uses the Moondream 0.5B model together with `AccessibilityService.takeScreenshot` and transient RAM-only frame processing to estimate whether a no-text item contributes negative exposure.

Video Dwell Time - The time interval spent on a viewed short-form video before the user swipes away or otherwise transitions, computed from heuristically inferred content-transition signals derived from accessibility events and used in this study as one behavioral proxy for scrolling engagement.

Very Large Online Platforms (VLOPs) - Large digital platforms, such as Facebook, Instagram, and TikTok, that operate at massive user scale and strongly shape online content exposure and engagement patterns (Chen et al., 2024).

Session Duration - The continuous period of active engagement with a target short-form video platform, as determined by interaction signals and the session-management rules described in Chapter 3.

Sentiment-Unreliable Session - A session in which item-level negativity cannot be derived reliably for the main analysis after the available text path or no-text VLM path has been attempted, such as sessions dominated by high-OOV text or by unresolved no-text items. Such sessions are excluded from the primary NSD- and DSI-based analyses.

Swipe/Scroll Transition Signals - Accessibility-detected navigation events used to heuristically infer when one short-form video ends and another begins, with text verification applied when usable caption text exists, enabling the computation of video dwell time and the logging of supporting interaction counts.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews peer-reviewed literature and empirical studies relevant to doomscrolling, short-form platform engagement, digital mindfulness, privacy-preserving mobile monitoring, sentiment analysis, and software evaluation. Journalistic, policy, platform-issued, and product-documentation materials are used only for contextual background or feature verification. The review is organized to clarify the study's evidence base, technical grounding, and remaining gaps.

2.1. Foreign Literatures

This section presents selected foreign literature relevant to doomscrolling, short-form platform design, digital mindfulness, and privacy-preserving mobile systems.

2.1.1. Doomscrolling, Platform Design, and Compulsive Engagement

Recent literature treats doomscrolling in short-form feeds as a joint product of user vulnerability and platform design rather than as a simple matter of screen-time volume. Rodrigues (2022) frames doomscrolling as compulsive information seeking that can intensify anxiety and helplessness, while Zenone et al. (2021) show that recommendation-driven environments can amplify harmful exposure and addictive-use patterns. Zhang (2023) complements that view by showing that TikTok is woven into users' routines as entertainment, information, and social reference, which helps explain why these feeds can sustain engagement so effectively.

These works suggest that time-only intervention logic is too coarse for the present problem. A defensible intervention system should monitor sustained interaction patterns within algorithmic short-form feeds rather than rely only on coarse daily time limits.

2.1.2. Digital Mindfulness and Technology-Assisted Self-Regulation

Recent digital well-being reviews converge on a narrow design implication: technology-assisted self-regulation support is more acceptable when it is brief, timely, and low-burden. Mitsea et al. (2023) provide limited support for digitally assisted mindfulness as one self-regulation strategy, while Antezana et al. (2022) show that engagement improves when interventions remain lightweight and relevant to the user's immediate context. Together, they support short adaptive prompts as low-burden, non-clinical interruption mechanisms.

2.1.3. Privacy, Persuasive Design, and Ethical Considerations in Digital Well-Being

Privacy-oriented digital well-being literature also constrains what a defensible system can do. Tewari (2023) argues that privacy protections should be designed into mobile health architectures from the start rather than added after collection logic is already fixed. That point is central to the present design: behavioral monitoring, sentiment processing, and prompting should occur without exporting raw user content to external services. This literature therefore supports local processing and aggregate-only retention.

2.1.4. Edge Computing and On-Device Vision-Language Fallbacks

Recent literature on privacy-preserving mobile analysis converges on a clear constraint: if visual inference is used at all, it should remain inside the device's privacy boundary. Sending screenshots to cloud-based multimodal APIs would expose sensitive user data to external servers. Sharshar et al. (2025) support the feasibility of edge-deployed Vision-Language Models under network and latency constraints, while Lee et al. (2024)

show why compact transformer-based vision models remain a practical design priority on mobile and edge devices. Broader multimodal sentiment and affective-computing reviews likewise show that visual channels can add information when text is absent, although interpretation remains limited (Cortiñas-Lorenzo & Lacey, 2024; Das & Singh, 2023; Johnson et al., 2025). This literature supports the app's on-device VLM fallback for no-text items, with Moondream 0.5B used as a small deployment-fit model.

2.2. Local Literatures

This section presents selected local literature and contextual materials relevant to doomscrolling, excessive social media use, and digital well-being in the Philippine setting. Because adult Philippine doomscrolling evidence remains limited, peer-reviewed local studies are used as nearby evidence of attention, scrolling, and self-regulation concerns, while journalistic or policy-oriented sources serve only as contextual background.

2.2.1. Local Discourse and Platform Concern

Philippine background materials show that doomscrolling and digital well-being had already entered local public discourse by 2021 Lanuza et al. (2021); Panaligan (2021).

2.2.2. Limits of the Local Evidence Base

The local empirical base remains uneven and still depends mainly on student-centered studies discussed later in this chapter. Recent Philippine work outside student-only samples is useful but indirect: Cleofas et al. (2022), Zamora et al. (2021), and Castillo et al. (2022) broaden the adult context by showing that social-media use intersects with mental well-being, anxiety-related susceptibility, or social connectedness across different Filipino populations. Even so, the local literature does not yet provide adult doomscrolling field evidence, validated cutoffs, or intervention-effect benchmarks for a privacy-preserving mobile system.

2.3. Interim Synthesis of Literature

The reviewed foreign and local literature establishes three points. First, doomscrolling is a recognizable behavioral construct linked to distress-related and attention-related concerns, and platform design appears to help sustain engagement (Hawwa et al., 2025; Qin et al., 2022; Taskin et al., 2024). Second, existing digital well-being responses remain fragmented: mainstream device- and platform-level tools still emphasize summaries, timers, or break reminders rather than user-controlled, locally processed, content-sensitive estimation (Apple Inc., 2025; Google, 2024; Mosseri, 2021; Rahmillah et al., 2023; Tewari, 2023; TikTok, 2024). Third, Philippine literature supports local relevance and some adult context, while adult field evidence on prevalence, thresholds, and intervention effects remains thin.

Together, these sources justify an engineering problem: combining behavioral signals, visible text, and a no-text VLM fallback in one privacy-preserving mobile system. Because text sentiment alone cannot represent multimodal short-form feeds, the study uses a hybrid exposure signal in which analyzable text is scored through the text path and no-text items are routed through the VLM fallback.

2.4. Foreign Studies

This section presents a review of related studies that provide empirical evidence on doomscrolling, compulsive social media use, and their psychological and behavioral impacts.

2.4.1. Psychometric Validation of Doomscrolling as a Measurable Construct

Sharma et al. (2022) establish doomscrolling as a measurable self-report construct, while Satici et al. (2023) extend that psychometric work and support a shorter 4-item form. Together, these studies justify using self-reported doomscrolling as a baseline convergent-association anchor rather than as session-level ground truth.

2.4.2. Psychological Mechanisms and Consequences of Doomscrolling

Taskin et al. (2024) and Hawwa et al. (2025) both position doomscrolling as part of a wider pathway linking platform engagement to poorer well-being outcomes. Taskin et al. (2024) highlight reduced mindfulness and secondary traumatic stress as mediating mechanisms, while Hawwa et al. (2025) connect doomscrolling to the relationship between social media addiction and anxiety. Their shared implication is that a pure time-only measure is insufficient.

2.4.3. Platform Addiction and Algorithmic Engagement

Qin et al. (2022) show that TikTok's platform characteristics can intensify flow experience and addictive use. This supports the thesis's focus on short-form mechanics and sustained engagement rather than on user weakness alone.

2.4.4. Computational Approaches to Detecting Harmful Digital Behavior

Zannettou et al. (2024) show that engagement with TikTok recommendation streams can be studied from donated behavioral traces, while Lokeshkumar et al. (2021) show that text and activity patterns can be processed computationally as risk-relevant indicators. These studies serve as methodological analogies that support the technical plausibility of trace-based behavioral analysis.

2.4.5. Fuzzy Logic as an Inference Engine for Mobile Behavioral Classification

Recent reviews show that fuzzy rule-based systems are useful when the target construct is gradual rather than binary and when the inference process must remain interpretable to human reviewers (Pickering et al., 2025; Vashishtha et al., 2023). Recent work on interpretable fuzzy-model design also continues to treat simple triangular or shoulder-style membership sets as practical, auditable choices when labeled session-level training data are not available (Khairuddin et al., 2021; Porebski, 2022). Recent fuzzy-design work also describes symmetric grid partitions and strong fuzzy partitions as interpretable starting structures when domain-specific numeric cutoffs are unavailable (Azam et al., 2021; Casalino et al., 2022; Khairuddin et al., 2021). Together, this literature supports using fuzzy logic and an interpretable starting structure for Negative Sentiment Density, where no universal percentage threshold set is established.

2.4.6. Data Extraction and Privacy-Preserving Edge Computing

Lee et al. (2022) show that Android Accessibility Service can expose fine-grained interaction data unavailable through standard APIs, but they also make clear that such access is privacy-sensitive. Swathi et al. (2025) complement that point by supporting edge-centric monitoring as a privacy-preserving and responsive architecture. Together, these studies justify the architectural stance that rich interaction data, if collected at all, should be processed locally and reduced to aggregate outputs rather than exported as raw content.

2.4.7. Social Media Use, Sustained Attention, and Digital Well-Being Interventions

Rajeswari & Vakkil (2024) and Roffarello & De Russis (2021) support different parts of the present design. Rajeswari & Vakkil (2024) link heavier social media engagement with poorer sustained attention, which supports the general relevance of prolonged use and viewing cadence as behavioral signals. Roffarello & De Russis (2021) show that smartphone habits can be monitored in real time and mitigated through proactive just-in-time reminders, which supports the feasibility of lightweight adaptive intervention on the device. Together, they justify attention to both continuous engagement patterns and in-the-moment interruption.

Recent measurement studies also help bound the fuzzy time-based inputs. Muise et al. (2024) show that smartphone content exposure often occurs in bursts of only a few seconds, Cho et al. (2021) treat sessions shorter than 5 seconds as likely mistaken openings and use 30 seconds as a short-duration divider in feature-level social-media logging, and Tian et al. (2021) report that 93.7% of mobile app usage in their large-scale logs lasts less than 10 minutes. Recent smartphone-logging work that reconstructs sessions from app-event streams also continues to merge adjacent app uses into one session when inter-app gaps remain within about 45 seconds (Ahmed & Ahmed, 2023; Chen et al., 2023). These studies justify distinguishing immediate skips, brief but nontrivial viewing, clearly sustained use, and very short app-switch interruptions instead of treating all exposure as one undifferentiated time scale.

Intervention-timing studies point in the same direction. Terzimehić & Aragon-Hahner (2022) report that sessions shorter than 10 minutes are less prone to regret, with alternative offline activities becoming more salient in the 10-20-minute and >20-minute ranges. Rixen et al. (2023) treat 10+ minute infinite-scrolling episodes as long sessions, Meinhardt et al. (2025) deploy an infinite-scroll intervention after 15 minutes of continuous scrolling, Ismail & Al Thani (2022) use personalized prompts after 40 minutes of continuous sitting, and Ikegaya et al. (2025) show that week-1-derived personalized intervention criteria outperform uniform criteria in the first hour after prompting. Recent JITAI reviews add the same design implication at the framework level: decision points and rules should be empirically grounded, burden-aware, and personalized when possible rather than fixed by one universal cutoff (Fiedler et al., 2024; Genugten et al., 2025; Hsu et al., 2025). Together, these studies support treating sub-10-minute use as a lower-risk baseline region, recognizing 15 minutes as a defensible live-prompt entry point for continuous infinite scrolling, and deriving Week 2 prompt criteria from each participant's Week 1 baseline.

2.4.8. Technology Acceptance Frameworks

Recent mHealth acceptability reviews continue to treat TAM as a practical evaluation framework, with Perceived Usefulness and Perceived Ease of Use retained as core acceptance constructs (Adnan et al., 2025). That supports the inclusion of TAM alongside ISO/IEC 25010 in the post-usage evaluation.

2.4.9. Computational Constraints in Code-Mixed Sentiment Analysis

Analyzing Taglish on mobile devices presents two challenges: computational load and high Out-of-Vocabulary (OOV) rates. Recent low-resource and code-mixed sentiment work shows that transformer approaches can be data-hungry or resource-intensive, while lexicon-based and hybrid methods remain relevant when reproducibility, low overhead, and resource scarcity matter (Hussain et al., 2025; Mohammed & Prasad, 2023; Nazir et al., 2026; Tho et al., 2021). Work on lexicon curation for low-resource social-media text also shows that slang, emoticons, and local usage patterns often require targeted valence tuning rather than direct reuse of generic sentiment resources (Hashmi et al., 2024; Khan et al., 2025; Perera & Caldera, 2024; Wijayanti & Arisal, 2021). For that reason, the study adopts a limited VADER + **Minimum Viable Lexicon** (MVL) strategy for code-mixed sentiment analysis on mobile devices.

2.4.10. Addressing the Affective Gap Through an On-Device VLM Fallback

When a short-form item contains no usable caption or visible comments, text-only sentiment cannot characterize that item at all. Reviews of multimodal sentiment analysis show why visual channels become relevant once text is absent (Das & Singh, 2023), while user-generated-video emotion studies show that keyframe-based reduction can support practical visual inference from video without requiring dense frame-by-frame processing (Wei et al., 2021; Zhang & Xu, 2023). Additional recent work shows that sparse visual sampling can remain useful under resource or privacy constraints, including sparse sampling with majority voting on short utterance clips (Sharma et al., 2023), privacy-compliant group emotion recognition using only 5 uniformly distributed frames on 5-second videos (Augusma et al., 2023), and sparse sampling regimes in recent VLM long-video evaluation under context-length constraints (Qu et al., 2025). These studies justify routing no-text items through the VLM fallback, using Moondream 0.5B as a small deployment-fit model with an on-demand screenshot capture rule rather than continuous video analysis.

2.5. Local Studies

2.5.1. Doomscrolling and Student Outcomes in the Philippines

Punzalan et al. (2024) provide the closest local study to the present topic by describing doomscrolling-related experiences among Filipino students. Bautista et al. (2024), Canila et al. (2023), Ababat et al. (2024), and Cleofas et al. (2022) reinforce the same local signal: heavier or more problematic social media use is linked to boredom proneness, distraction, procrastination, attention-related strain, or poorer mental well-being among Filipino students and young adults. These studies establish Philippine relevance but remain student-centered, self-report-heavy, or qualitative, supporting cautious adult field evaluation.

2.5.2. Social Media, Reading Competence, and Attention Span

Gagalang (2021) and Cardoso et al. (2024) together support the local relevance of attention-related harms associated with heavy digital-platform use, especially among students. They help explain why viewing cadence and sustained use are reasonable behavioral dimensions to observe.

2.5.3. Cognitive Benefits of Digital Detox Interventions

Lim et al. (2025) show that a two-week digital detox can improve attention and memory among Filipino Grade 12 students. The study is relevant because it makes a two-week observation window plausible.

2.5.4. Local Computational Work on Filipino Sentiment Analysis

Co et al. (2022) and Cruz & Balahadia (2022) show that Filipino or English-Filipino public text can be analyzed with lexicon-based or lightweight sentiment pipelines, but they also make clear that domain adaptation remains necessary when language use shifts across settings. These studies show that Filipino-facing sentiment processing is technically workable in the local setting when the lexicon is treated as task-specific.

2.5.5. Software Quality Evaluation in Philippine Educational Technology

Software evaluation is anchored on the official ISO/IEC 25010 product quality model (International Organization for Standardization, 2023). The standard supports the choice of selected quality characteristics, while the survey wording used in Chapter 3 is researcher-developed and subject to expert review.

2.6. Integrated Synthesis and Research Gap

2.6.1. Comparative Analysis of Existing Digital Well-Being Tools

To contextualize the research gap identified in this review, the following table summarizes the capabilities of existing digital well-being tools and platform-level features. The comparison is based on five key requirements derived from the preceding literature.

Feature characterizations for Apple Screen Time, Google Digital Wellbeing, TikTok, and Instagram are drawn from their official product or support pages (Apple Inc., 2025; Google, 2024; Mosseri, 2021; TikTok, 2024). The broader third-party app row is grounded in the multimethod review by Rahmillah et al. (2023).

Operational Definitions for Comparison:

- **Behavioral Monitoring:** [Real-time] = continuous background monitoring of specific actions; [Aggregate] = summary views or timer-based monitoring without item-level sensing; [Platform-only] = monitoring limited to one platform's own activity tools.
- **Text-Visible Sentiment Proxy:** [✓] = analyzes the emotional tone of extracted text content; [✕] = treats all viewed content equally regardless of text tone.
- **Adaptive Intervention:** [✓] = intervention type changes based on estimated severity; [Partial] = fixed caps, timers, or break nudges that do not estimate content or risk; [✕] = monitoring only or fixed reminders without severity adaptation.
- **On-Device Processing:** [✓] = core monitoring or intervention logic is user-controlled and executed locally on the device; [N/A] = feature is platform-managed and not exposed as a user-controlled local processing pipeline.

- **Platform Scope:** [System-wide] = works across many apps on the device; [Single-platform] = limited to one service; [Varies] = depends on the specific third-party app; [Specific targets] = limited to named study platforms.

Table 1

Comparative Analysis of Digital Well-Being Tools and the Proposed System

Tool / System	Behavioral Monitoring	Text-Visible Sentiment Proxy	Adaptive Intervention	On-Device Processing	Platform Scope
Apple Screen Time	Aggregate	×	Partial	✓	System-wide
Google Digital Wellbeing	Aggregate	×	Partial	✓	System-wide
TikTok screen-time tools	Platform-only	×	Partial	N/A	Single-platform: TikTok
Instagram daily limit / break reminders	Platform-only	×	×	N/A	Single-platform: Instagram
Rahmillah et al. (2023) apps	Varies	×	Partial	Varies	Varies
Proposed System	**Real-time**	✓	✓	✓	**Specific targets: TikTok, Facebook Reels, Instagram Reels**

As shown in the table, the reviewed literature and official product pages identify useful summaries, timers, and break mechanisms, but not a single adult short-form video system that combines real-time behavioral monitoring, a text-first sentiment proxy, an implemented no-text VLM fallback, adaptive prompting, and strict local processing. The gap therefore lies in integration and deployment scope: the ingredients appear separately in prior work, but not as one privacy-preserving artifact under the conditions targeted here. Buchanan et al. (2021) help motivate this search by showing that scrolling outcomes depend on exposure quality rather than time alone, while Roffarello & De Russis (2021) show that just-in-time mobile interventions are feasible.

The reviewed studies converge on three themes. First, doomscrolling can be measured through self-report and discussed as a distinct construct, which supports a baseline convergent association check (Satici et al., 2023; Sharma et al., 2022). Second, technical and intervention precedents exist for mixed mobile interventions, interpretable fuzzy inference, privacy-preserving on-device processing, Filipino-facing sentiment analysis, and narrow no-text visual inference on edge devices, although the closest studies remain analogies rather than direct implementations of the app evaluated here (Co et al., 2022; Cruz & Balahadia, 2022; Das & Singh, 2023; Lee et al., 2022; Pickering et al., 2025; Roffarello & De Russis, 2021; Sharshar et al., 2025; Vashishtha et al., 2023; Wei et al., 2021). Third, the local Philippine literature indicates that attention, self-regulation, and problematic-use concerns are locally relevant, that some adult social-media contexts have been studied, that short-term digital well-being observation windows are workable, and that Filipino-facing sentiment processing is technically feasible in local text settings (Castillo et al., 2022; Cleofas et al., 2022; Co et al., 2022; Cruz & Balahadia, 2022; Lim et al., 2025; Punzalan et al., 2024; Zamora et al., 2021).

A clear **research gap** therefore remains. No reviewed study integrates behavioral logging, a text-first sentiment proxy, an implemented no-text VLM fallback, adaptive prompts, and strict local processing in one privacy-preserving adult short-form video system while also reporting a field evaluation aligned with this thesis. The local adult evidence base is also too limited to justify calibrated estimator thresholds, strong expected-effect claims, or clinical interpretation. Hybrid text-plus-visual sentiment routing is therefore treated as partial coverage of negative exposure. In this study, NSD is defined as a session-level negative-exposure metric computed from resolvable text units and VLM-resolved no-text items. The study addresses these gaps through a two-week pilot field evaluation focused on software behavior, short-term behavioral comparison, user acceptance, expert review, baseline convergent association, and sentiment-reliable coverage reporting.

Chapter 3

METHODOLOGY

This chapter presents the methodology used for designing, developing, and evaluating the system. It covers the research design, architecture, development procedure, evaluation method, data collection, respondents, and statistical treatment.

3.1. Type of Research

This study adopted a **design-and-development research** approach with a **two-group baseline-intervention pilot field evaluation** (Creswell & Creswell, 2022). The design-and-development component focused on building the software artifact and its heuristic risk-estimation logic, while the field-evaluation component observed the system under real-world conditions.

The study followed a primarily quantitative orientation, with open-ended feedback used as supplementary qualitative context. Automated system logs and structured survey responses were used to assess software quality, user acceptance, expert review, baseline convergent association, and short-term behavioral differences after data collection.

3.2. Research Design

The study used a **two-group baseline-intervention field design** treated as a **pilot-scale randomized field evaluation**. After consent and study-compatibility screening, participants were assigned 1:1 to either an **intervention group** or a **logging-only control group** using a concealed permuted-block allocation list with randomly ordered block sizes of 2 and 4. The allocation list was generated before enrollment by a researcher who was not responsible for participant screening and released only after eligibility was confirmed, which helped preserve balance without making the next assignment easily predictable. Participants then completed an identical one-week baseline logging phase with prompts disabled. During Week 2, the intervention group received adaptive prompts while the control group continued logging-only.

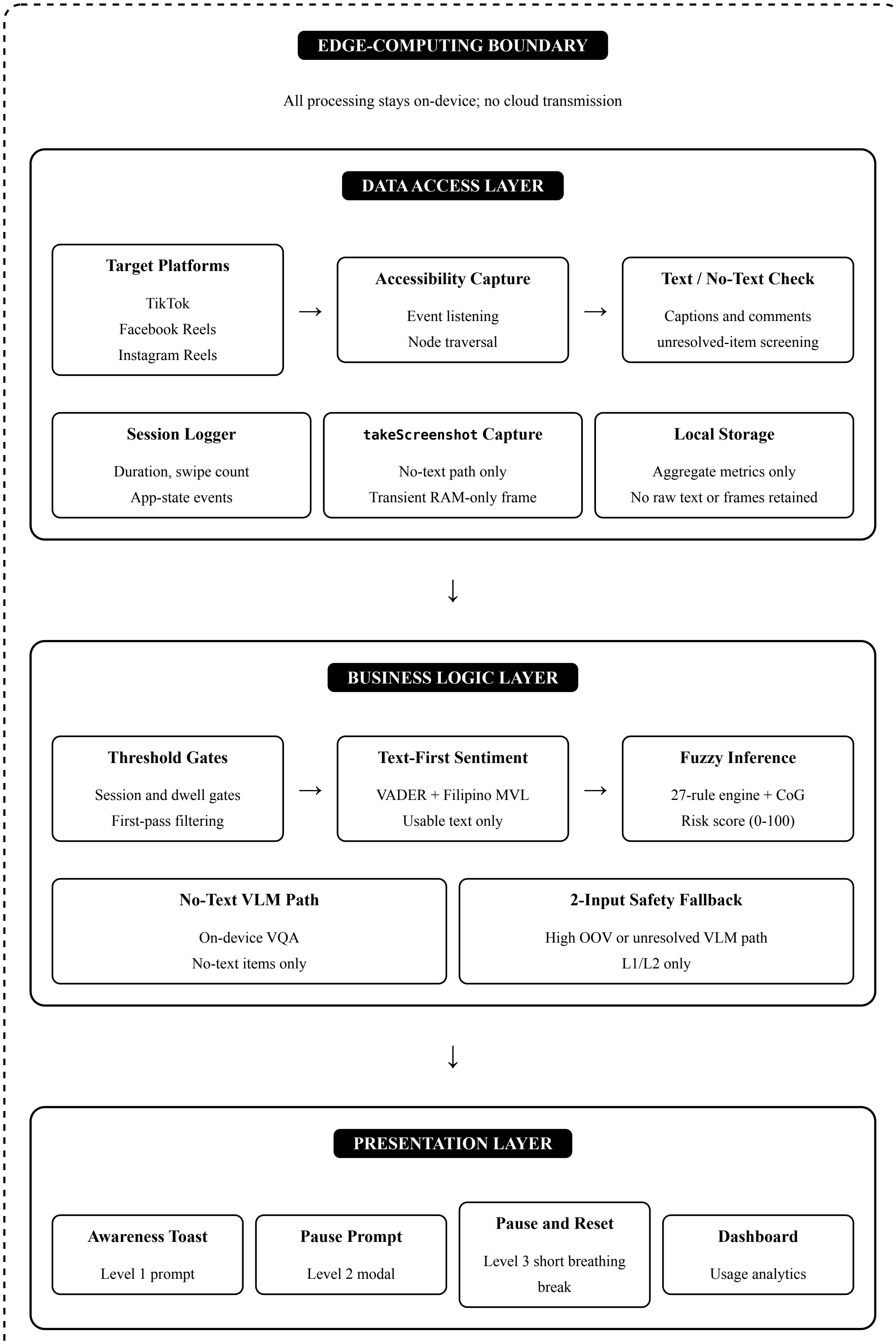
The study questions were answered through related but distinct methodological parts. Research Question 1 is addressed by the architecture, routing, variable-justification, and algorithm sections. Research Question 2 is addressed by Week 1-to-Week 2 change analyses of selected logged metrics and self-reported doomscrolling scores, comparing the intervention and control groups and using a paired within-intervention comparison as secondary evidence. As a pilot-scale study, these analyses estimate short-term observed differences under the study conditions rather than definitive intervention efficacy. Research Question 3 is addressed by the baseline convergent association between Week 1 DSI and the Doomscrolling Scale. Research Questions 4 and 5 are addressed by the ISO/IEC 25010, TAM, and SME evaluations.

3.3. System Architecture

The system uses a **modular three-layer architecture** to support maintainability, interpretability, and separation of concerns. Its design follows **Privacy by Design**, keeping processing local to the device through an edge-computing approach. The application is organized into the **Data Access Layer**, **Business Logic Layer**, and **Presentation Layer**, allowing data collection, estimation logic, and user-facing components to be developed with minimal coupling. The architecture also avoids device rooting by relying on Android Accessibility Service for text extraction, interaction-event monitoring, and screenshot-assisted VLM fallback when no usable caption or visible comments exist. The figure below shows this hybrid text/VLM pipeline.

Figure 3

System Architecture Diagram: Three-Layer Edge Computing Design



3.3.1. Data Extraction Strategy

The primary mechanism for data acquisition is the Android **Accessibility Service API**. It lets the system read captions, comments, and interaction events from short-form video apps without requiring device rooting. The same layer also captures the interaction signals needed for session duration, content-transition detection, video dwell-time computation, and supporting swipe logs. When a viewed item has no usable caption or visible comments, the enabled accessibility service uses `AccessibilityService.takeScreenshot` to capture an on-demand transient screen frame for the no-text VLM fallback.

3.3.2. Accessibility Service Implementation

The implementation entails creating a background service that extends the `AccessibilityService` class. Functionally, this allows the system to observe the UI hierarchy of other apps in real-time, effectively perceiving the screen content as a screen reader would for accessibility purposes.

The technical implementation involves the following core components:

- **Service Extension**: Development of a custom service inheriting from `AccessibilityService`.
- **Event Listening**: The service subscribes to specific system events, primarily `AccessibilityEvent.TYPE_WINDOW_CONTENT_CHANGED` and `TYPE_VIEW_SCROLLED`, to detect dynamic content updates.
- **Node Traversal**: Upon capturing an event, the system traverses the hierarchy of `AccessibilityNodeInfo` objects, filtering specifically for `TextView` elements to extract creator captions and any visible user comments already present on-screen as affective reaction signals. The system does not actively open hidden comment panels or trigger additional UI states during extraction.

- **Scroll-Event Disambiguation**: Because `TYPE_VIEW_SCROLLED` events are generated by both feed-level video transitions and within-video comment-panel scrolling, the system does not treat every scroll event as a new item by default. When usable caption text is present, the system applies a content-change verification step before registering a content transition: after each feed-level scroll event, the caption text is re-extracted and compared to the previously recorded caption using exact string equality after whitespace normalization, and a transition is registered only if the caption has changed. When a viewed item has no usable caption or visible comments, however, the system cannot rely on caption comparison; in that no-text case, a feed-level scroll event is treated as a heuristic new-item boundary unless suppressed by comment-panel state or similar non-feed interaction context. As a secondary check, the system tracks comment-panel open/close state via `TYPE_WINDOW_STATE_CHANGED` to suppress transition processing during comment browsing. Comment-panel scrolling, caption expansion, and UI-only animations therefore do not reset the dwell-time interval or increment the swipe count. Comments visible during panel scrolling are still extracted for sentiment analysis and associated with the current video item.

3.3.3. Constraints and Challenges

A significant challenge in this architecture is the requirement to reverse-engineer the UI structure of targeted applications. The extraction logic must identify specific resource IDs (e.g., `com.zhiliaoapp.musically:id/caption_text`) to distinguish meaningful content from UI clutter such as buttons or timestamps. This approach has inherent fragility: changes to the target application's UI structure or IDs by their respective developers may disrupt the scraper's functionality, representing a known limitation of this methodology.

3.3.4. Edge Computing and Privacy Architecture

The system follows edge computing principles to mitigate privacy risks:

1. **No Persistent Storage of Raw Content**: In the study version, scraped text content and temporary no-text screen frames are processed entirely in RAM and immediately discarded after scoring. Only derived numeric scores, aggregate usage metrics, and local configuration data are retained on-device.
2. **On-Device Processing**: All Fuzzy Logic inference, VADER analysis, and no-text VLM inference occur locally on the user's device. No content is transmitted to external servers.
3. **User-Initiated Consent**: The Accessibility Service requires explicit user activation through Android system settings. The same enabled service provides the no-text screenshot capability through `android:canTakeScreenshot="true"` and `AccessibilityService.takeScreenshot`, so the study's operational data-access consent is centered on the Accessibility Service setup.
4. **Aggregate-Only History**: The usage dashboard displays only aggregate statistics (e.g., "negative content exposure: 45%") rather than storing individual text samples or captured frames.

3.3.5. Extraction Failure Handling

This architecture relies on hardcoded UI resource IDs (e.g., `com.zhiliaoapp.musically:id/caption_text`) within the Accessibility Service. Mid-experiment UI updates can restructure the target application's view hierarchy and disrupt extraction. Such breakage is logged as a **reliability event**, and sessions or participant-periods that cannot yield valid extraction output are excluded from the analyses that require those measures.

3.3.6. Mindfulness Intervention System

Based on digital well-being and smartphone-habit intervention literature, including recent just-in-time adaptive intervention work on timing, feasibility, and burden management (Ismail & Al Thani, 2022; Mair et al., 2022; Roffarello & De Russis, 2021; Teepe et al., 2021; Wang et al., 2023; Yang et al., 2023), the system uses a multi-level adaptive intervention approach:

Level 1 - Awareness Notification: A non-intrusive toast message appears when the RiskScore enters the lower Warning band, informing the user of their session duration.

Level 2 - Pause Prompt: A modal dialog offers the user options to continue, take a break, or view their usage statistics when the RiskScore enters the upper Warning band.

Level 3 - Pause and Reset / Short Breathing Break: A full-screen overlay presents a brief guided breathing animation (study default: 60 seconds) when the RiskScore enters the Critical band. Following the psychology expert's recommendation, this feature is framed as a non-clinical pause-and-reset prompt rather than as a therapeutic intervention (Mitsea et al., 2023; Yang et al., 2023).

Across all three levels, prompt wording is designed to remain calm, supportive, and non-judgmental. User-facing language avoids diagnostic labels such as "risk detected," "critical behavior," or "you are doomscrolling." The implementation also preserves user autonomy by allowing users to dismiss prompts, continue use, take a break, or view usage information. Recommended safeguards for future iterations include explicit skip and disable controls, configurable prompt sensitivity, and clearer onboarding or dashboard disclaimers that explain the app provides heuristic pattern monitoring only.

For analytic transparency, time spent inside system-generated modals or overlays is excluded from prompt-excluded active-use metrics. When an intervention appears, the ongoing content-view interval is closed before the prompt begins, and a new interval starts only if the user returns to the target platform afterward. Raw elapsed session time and raw elapsed dwell time are retained separately for the primary behavioral comparison in the evaluation phase.

Usage Dashboard: Shows aggregated usage and sentiment trends over time to support self-reflection and behavioral awareness. It is treated as a secondary reflective feature rather than as the primary intervention mechanism.

3.3.7. Design Justification for Risk-Estimation Variables

This section addresses Research Question 1 by mapping each risk-estimation variable to the Doomscrolling Feedback Loop Model (Sharma et al., 2022) and to the reviewed literature. Variables were retained only if they were conceptually plausible, literature-supported, and observable through Android without rooting.

Table 2

Design Justification Matrix: Theoretical Constructs to System Variables

Theoretical Construct	Observable Proxy	System Variable	Data Source	Unit
Behavior: Persistent, obsessive consumption	Prolonged or rapid navigation through content without meaningful pauses	Video Dwell Time	AccessibilityService (active time between TYPE_VIEW_SCROLLED events, gated by micro-interaction events)	seconds
Behavior: Repetitive rapid feed navigation	Frequency of moving through items during a session	Swipe Count (supporting interaction log)	AccessibilityService view-scroll event stream	count
Behavior: Prolonged maladaptive engagement	Extended time on target apps without voluntary breaks	Session Duration	Accessibility-derived session tracker and app-state events	minutes
Antecedents/ Triggers: Negative content exposure	Proportion of consumed content with negative emotional tone	Negative Sentiment Density (NSD)	Negative-exposure density from analyzable text units and VLM-resolved no-text items	% (0–100)

Video Dwell Time was selected because TikTok and Reels use a discrete, paginated swipe interface rather than continuous scrolling. Dwell time - measured as the duration in seconds between consecutive content-verified transition events — captures how quickly or slowly users move through content. Because the Accessibility Service cannot reliably read exact platform-reported video duration across all target applications, dwell time is treated as a behavioral proxy. **Swipe Count** is retained only as a supporting interaction metric that helps show whether a long session consists of rapid chaining or slower browsing.

Session Duration is operationalized as continuous active time on a target application, reflecting the prolonged-engagement dimension reported in doomscrolling-related literature. Together, dwell time and session duration describe how long and how continuously the user remains engaged.

Negative Sentiment Density (NSD) captures the negative-exposure dimension of doomscrolling through a hybrid operationalization. When usable text exists, analyzable caption and visible-comment units are scored through the text path. When no usable text exists, the item is routed through the VLM fallback and contributes one resolved no-text label to the same session-level exposure estimate. High-OOV text is not rerouted into the VLM; if negativity still cannot be resolved reliably, the session enters the 2-input behavioral fallback and is marked sentiment-unreliable for the main sentiment-based analyses.

NSD is therefore an exposure proxy, not a direct measure of distress.

Session Boundary Definition: A session begins when a target app enters the foreground (detected via `TYPE_WINDOW_STATE_CHANGED`) and ends when: (a) the target app leaves the foreground for longer than a 30-second bridge window, (b) the screen turns off, or (c) the Accessibility Service is terminated. If the user briefly switches away and returns within 30 seconds, the gap is excluded from duration and dwell-time accumulation but the session continues. Recent smartphone-logging studies that reconstruct sessions from app-event streams continue to use 45-second inter-app gaps when direct lock/unlock boundaries are unavailable (Ahmed & Ahmed, 2023; Chen et al., 2023). The present system therefore treats 30 seconds as a stricter study-defined session-continuity heuristic within that recent measurement range so only very brief app switches are merged; longer away intervals are treated as new sessions. The reproducibility notes treat ±15-second perturbations as the relevant local check for this fixed study prior.

Idle Timeout Heuristic (Active-vs-Idle Disambiguation): To prevent false inflation of dwell time when the device is unattended (e.g., phone left on a looping video), the dwell accumulator is treated as **active** only when periodic micro-interactions are observed (e.g., `TYPE_VIEW_CLICKED`, `TYPE_TOUCH_INTERACTION_START/END`, or equivalent interaction signals exposed by the Accessibility layer). If no micro-interaction occurs within a rolling 45-second window, the ongoing dwell interval is paused and marked as idle exposure until interaction resumes or the next swipe occurs. Recent smartphone-session reconstruction studies continue to treat gaps of about 45 seconds as a plausible continuity delimiter when direct boundary data are unavailable (Ahmed & Ahmed, 2023; Chen et al., 2023). The dwell tracker uses the same order-of-magnitude value as a study-defined inactivity cap to distinguish active viewing from unattended looping: if interaction signals disappear beyond that range, accumulation pauses until interaction resumes or the next swipe occurs.

$$\text{NSD} = \left(\frac{\text{Negative resolvable units}}{\text{Resolvable units in session}}\right) \times 100$$

To formally represent this calculation in mathematical notation:

$$\text{NSD} = \frac{\sum_{i=1}^{N_R} \mathbb{I}(y_i = 1)}{N_R} \times 100$$

Where N_R is the number of session units whose negativity can be resolved either from analyzable text or from the no-text VLM path, and y_i is the binary negative label assigned to resolved unit i. For text-resolved units, negativity follows the VADER criterion $C_i < -0.05$. For no-text VLM-resolved units, negativity is assigned when the returned visual label is `SEVERE_NEG` or `MILD_NEG`. The indicator function $\mathbb{I}$ assigns 1 only to resolved units classified as negative and 0 otherwise. This makes the reported NSD a conservative negative-item density rather than a continuous magnitude-weighted affect score.

The −0.05 threshold follows VADER's standard classification convention, where compound scores below −0.05 are treated as negative sentiment (Hutto & Gilbert, 2014). Caption/comment text units excluded by the language-reliability screen are removed from the text-side evaluation. No-text items routed to the VLM contribute one resolved item-level label each: `SEVERE_NEG` and `MILD_NEG` count as negative exposure, while `NEUTRAL`, `MILD_POS`, and `SEVERE_POS` count as resolved non-negative exposure. Sessions with very few resolvable units are flagged descriptively during analysis.

Alternative candidates such as swipe-back frequency, time-of-day patterns, and biometric signals were excluded because they either lack enough empirical support in the doomscrolling literature or are technically infeasible within the intended Android monitoring environment. Likewise, although the **internal psychological Antecedents** of the Feedback Loop Model (e.g., anxiety, FoMO) remain conceptually important, they are not measured directly because real-time capture would require intrusive experience sampling or biometric sensors. The system therefore estimates doomscrolling-related risk from one exposure proxy, NSD, and two behavioral proxies, Video Dwell Time and Session Duration.

3.3.8. Analytic Scoring Framework

Week-level DSI is computed on one fixed scoring framework across both study weeks so the analytic scale remains constant. Swipe events serve only as transition markers for dwell-time computation, and prompt-response logs are reported descriptively because they exist only in Week 2. Live prompting may personalize Session Duration and NSD memberships after Week 1, but analytic DSI scoring remains fixed across both weeks.

3.4. Hardware and Software Specifications

Development uses Android Studio (Ladybug 2024.2.1+) with Kotlin 1.9+ on a machine with at least 8 GB RAM. The deployable application targets Android 8.0 (API

26) or higher, while the screenshot-assisted no-text VLM path requires Android 11 (API 30) or higher because it uses `AccessibilityService.takeScreenshot`. Local storage uses Room (SQLite), sentiment analysis uses a VADER-compatible lexicon-and-rule engine re-implemented in Kotlin for on-device Android deployment, dashboard visualization uses `MPAndroidChart`, screen-content extraction uses `androidx.accessibilityservice`, and no-text routing uses the Android Accessibility Service together with the on-device Moondream 0.5B VLM/VQA module. A physical Android device is used for field testing.

3.5. Method in Developing Software

3.5.1. Methodology Procedure

3.5.1.1. Agile Scrum

Development follows **Agile Scrum** in four iterative one-week sprints covering: (1) requirements finalization and planning, (2) accessibility extraction and local logging, (3) estimator and prompting integration, and (4) testing, refinement, and deployment preparation. Scrum is selected because the Accessibility Service extraction logic requires frequent adaptation as target-app UI structures change, and the fuzzy-logic rule base requires iterative tuning.

3.5.2. Algorithm

The core logic of the system combines **Threshold-Based Rules**, **VADER (Valence Aware Dictionary and sEntiment Reasoner)**, the no-text VLM fallback, and **Fuzzy Logic**. Together, these components govern live prompting, session-level scoring, and week-level DSI computation.

1. Threshold-Based Heuristics: For **live prompting**, the system first applies duration-based gates before deeper inference. Recent JITAI and infinite-scroll studies support burden-aware timing and treat sub-10-minute use as relatively brief while placing sustained scrolling in the 10 to 20+ minute range (Fiedler et al., 2024; Genugten et al., 2025; Hsu et al., 2025; Meinhardt et al., 2025; Rixen et al., 2023; Teepe et al., 2021; Terzimehić & Aragon-Hahner, 2022; Wang et al., 2023). Accordingly, the system uses 15 minutes as the study-defined live-prompt gate and cooldown interval.

- **Session Duration Gate:** deeper prompting begins after 15 minutes of continuous use.
- **Intervention Cooldown:** after a prompt is shown, the system requires another 15 minutes of qualifying use before a new prompt can be triggered.

Mean Video Dwell Time contributes to the fuzzy engine once prompt eligibility is met.

Logged session summaries are retained whether or not a live prompt occurs.

2. VADER Sentiment Analysis: VADER is a lexicon- and rule-based sentiment analysis tool designed for social-media text and suited to on-device use (Hutto & Gilbert, 2014).

- **Input:** Extracted caption or comment text (e.g., “This is heartbreaking 😭”).
- **Process:** Tokenization and lexicon lookup.
- **Output:** A Compound Score ranging from −1.0 (Most Negative) to +1.0 (Most Positive). Mathematically, VADER computes this bounded score via the normalization scalar:

$$C = \frac{x}{\sqrt{x^2 + \alpha}}$$

Where x is the sum of the valence scores of recognized words, and α is the standard normalization constant ($\alpha = 15$).

The implementation also retains VADER’s native UTF-8 emoji valence parsing, so emoji-dominant comments (e.g., “😭😭” or “💀💀💀”) still contribute affective signal even when lexical Taglish syntax is highly out-of-vocabulary.

Filipino Lexicon Extension: To handle Filipino and Taglish expressions, the system extends VADER with a limited **Minimum Viable Lexicon** (MVL). Recent low-resource sentiment work and lexicon-curation studies support this approach for code-mixed, resource-constrained settings (Mohammed & Prasad, 2023; Nazir et al., 2026; Tho et al., 2021; Wijayanti & Arisal, 2021).

1. **Corpus Collection:** Before participant deployment, a small set of high-frequency non-English tokens is compiled from exploratory extraction logs produced during researcher-controlled test sessions or earliest non-study dry runs. The Filipino review instrument contained 53 MVL candidate terms, and the runtime lexicon preserves those terms together with four common/formal spelling counterparts for recurrent `naka-` forms. This study-defined scope keeps the lexicon focused on dominant recurrent tokens.
2. **Expert Polarity Rating Instrument:** The collected tokens are prepared in the Filipino MVL review instrument, where a Filipino language expert is asked to rate each token on a sentiment polarity scale from −4 (most negative) to +4 (most positive), following VADER's valence-rating convention. The instrument was completed by one expert reviewer, a doctorate holder in Filipino-language education, who rated all 53 candidate terms and confirmed the proposed neutral stop-word valences; the completed instrument and signed validation certificate are retained in the study's expert-review records.
3. **Single-Expert Review and Concordance Check:** Because the review relied on a single language expert rather than a multi-rater panel, no inter-rater agreement statistic can be computed, and this is stated as a study limitation; recent sentiment-annotation work shows that valence agreement is difficult even with explicit instructions and multiple raters (Krušic, 2024; Äyräväinen et al., 2025). As a transparency safeguard, the expert's ratings were compared post hoc against the valence values deployed in the runtime lexicon: all 53 expert ratings matched

the deployed values exactly (53/53 exact concordance), and all proposed stop words were confirmed as sentiment-neutral.

4. **Integration:** The runtime implementation appends Filipino/Taglish MVL terms and their implementation valence values to the VADER lexicon dictionary. The current runtime lexicon contains 57 Filipino/Taglish MVL entries: 56 affective entries and one neutral candidate entry (`buhay`, 0.0). It includes the review-instrument candidate terms and common/formal `naka-` spelling counterparts used in the Android implementation, such as `nakagagalit`/`nakakagalit`, `nakaiinis`/`nakakainis`, `nakalulungkot`/`nakakalungkot`, and `nakatatakot`/`nakakatakot`.
5. **Neutral OOV-Reduction Terms:** The Filipino MVL review instrument also lists Filipino stop words proposed for 0.0 valence. These terms are counted as recognized tokens to reduce false out-of-vocabulary inflation, but they do not add positive or negative sentiment. The complete runtime MVL and neutral OOV-reduction inventories are documented in Appendix D.

This approach keeps the vocabulary focused on recurring tokens while preserving annotation quality and computational efficiency for edge deployment.

3. Taglish Syntax Handling and Fallback Logic: The system uses a lightweight lexicon-based approach to keep inference on-device under mobile resource and privacy constraints. Local and low-resource studies support this choice for Philippine social-media text and code-mixed sentiment analysis (Co et al., 2022; Cruz & Balahadia, 2022; Mohammed & Prasad, 2023; Nazir et al., 2026; Pacol & Palaoag, 2021).

To address negations such as “hindi maganda” (“not good”), the system implements a localized negation heuristic. A predefined list of common Filipino negation markers (e.g., “hindi”, “di”, “wala”, “wag”) is integrated into VADER’s pre-processing step. If a negation marker is detected within three tokens preceding a recognized sentiment token, the polarity of that token is inverted before the final compound score is computed.

Informal social-media text also contains slang, textual variation, and deliberate misspellings. Unmatched variants remain unmatched and contribute to the OOV calculation rather than being force-mapped into a sentiment label through fuzzy matching. The neutral Filipino OOV-reduction terms only reduce false OOV counts for high-frequency non-affective function words; they do not change the compound sentiment score.

Dialect-heavy text and purely visual items can still fall outside the reliable text path. The system addresses that case through the no-text VLM fallback:

3.5.3. No-Text Sentiment Fallback via Moondream 0.5B

Short-form video environments can include purely visual items without accompanying text payloads. For that case, the system uses a **Per-Video Routing Heuristic** that calls a VLM fallback built on Moondream 0.5B. Moondream 0.5B is used as a small deployment-fit VLM for on-device edge inference (Lee et al., 2024; Sharshar et al., 2025).

- **Dynamic Routing Logic:** When transitioning to a new item in the feed, if the Accessibility Service successfully extracts usable text payload (caption or visible comments), the item is evaluated via VADER and routed into the text calculation tier. If no usable text is present (e.g., purely visual content without captions), the item enters the Accessibility Service screenshot plus Moondream 0.5B path.
- **Data Extraction via `AccessibilityService.takeScreenshot`:** Visually routed items use an on-demand screenshot after the new item has settled. This is a resource-aware compromise supported by related affective-video and privacy-compliant video analysis work showing that sparse visual evidence can support practical inference under resource constraints (Augusma et al., 2023; Qu et al., 2025; Sharma et al., 2023). To maintain the edge-computing privacy boundary, the captured frame exists only in volatile RAM for VLM inference, ensuring no user screen data is committed to persistent storage.

- **VQA Zero-Shot Computation:** The transient screenshot is processed entirely on-device by the Moondream 0.5B VLM/VQA module. The system uses constrained Visual Question Answering (VQA), and the no-text item is assigned one of five labels: `SEVERE_NEG`, `MILD_NEG`, `NEUTRAL`, `MILD_POS`, or `SEVERE_POS`. If the label is `SEVERE_NEG` or `MILD_NEG`, the item contributes one negative resolved unit to NSD; otherwise it contributes one resolved non-negative unit.

This route extends NSD coverage to no-text items while remaining within the device's privacy boundary. It remains an item-level visual classification path rather than a full multimodal interpretation of viewed media (Cortiñas-Lorenzo & Lacey, 2024; Das & Singh, 2023; Johnson et al., 2025; Sharshar et al., 2025; Wei et al., 2021; Zhang & Xu, 2023).

Safety Lock: If the no-text VLM path fails because `AccessibilityService.takeScreenshot` is unavailable, a screenshot cannot be captured, or VLM inference does not produce a stable item label, the system drops to 2-input fuzzy inference (Video Dwell Time + Session Duration only). Sessions dominated by high-OOV text follow the same fallback path. In either case, unresolved sessions are marked **Sentiment-Unreliable** and excluded from main Doomscroll Severity Index (DSI) statistical evaluation. The degraded rule base and prompt-mapping thresholds are detailed in Appendix B.

Operationally, each viewed item stays on the text path when usable text exists and moves to the VLM fallback only when no usable text exists. If the session-level OOV ratio reaches 50% or more, or if no-text resolution fails, the session is marked **Sentiment-Unreliable**: the sentiment input is dropped, inference proceeds on Video Dwell Time and Session Duration only, Level 3 prompts are disabled, and the session is excluded from main DSI computation. The 50% OOV screen functions as a study-defined reliability rule, and the study reports the number and proportion of sentiment-unreliable sessions to show DSI coverage.

4. Fuzzy Logic Inference System: The system uses fuzzy logic because the target construct is gradual, the rule base must remain inspectable for SME review, and labeled session data are not available for supervised on-device modeling (Pickering et al., 2025; Vashishtha et al., 2023). The fuzzy engine operates at the per-session inference level, while the statistical tests in Chapter 4 are used later for sample-level evaluation. The membership boundaries below define the fixed scoring framework for week-level DSI across both study weeks. In Week 2 live prompting, Session Duration and NSD memberships may be personalized from Week 1 sentiment-reliable sessions, while Video Dwell Time remains fixed (Ikegaya et al., 2025).

- **Fuzzification:** Crisp inputs (Video Dwell Time in s, Negative Sentiment Density in %, Session Duration in min) are converted into fuzzy sets using triangular membership functions. Mathematically, a triangular membership function with bounds (a, b, c) is defined as:

$$\mu_A(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a < x < b \\ \frac{c-x}{c-b} & \text{if } b \leq x < c \\ 0 & \text{if } x \geq c \end{cases}$$

With three input variables and three linguistic states per variable (Low, Medium, High), the rule base contains $3^3 = 27$ rules. The table below presents the membership boundaries used for analytic scoring:

Table 3

Fuzzy Set Membership Boundaries for Analytic Scoring

Variable	Low	Medium	High
Video Dwell Time (s)	0–5	4–20	15–30+
Negative Sentiment Density (%)	0–33	17–83	67–100
Session Duration (min)	0–10	8–20	15–40+

These boundaries combine recent app-use timing evidence with a regular three-part partition for NSD, where the literature does not provide standard per-session negativity thresholds (Azam et al., 2021; Casalino et al., 2022; Cho et al., 2021; Ismail & Al Thani, 2022; Khairuddin et al., 2021; Meinhardt et al., 2025; Muise et al., 2024; Terzimehić &

Aragon-Hahner, 2022; Tian et al., 2021). They serve as the pilot scoring structure for this estimator. Session Duration and Negative Sentiment Density act as the main risk axes, while Video Dwell Time mainly adjusts risk by showing how sustained attention is distributed across viewed items.

The table reports support intervals rather than every triangular parameter. In implementation, each Low set is a left-shoulder triangle anchored at zero, each Medium set peaks at the midpoint of its displayed support interval (12 s, 50%, and 15 min), and each High set is a right-shoulder triangle that saturates at the upper cap (30 s, 100%, and 40 min). This setup was chosen because it is the simplest piecewise-linear form that matches the declared Low/Medium/High meanings, keeps adjacent overlap without gaps, prevents overlap between non-adjacent sets, and keeps every parameter easy to check against the table.

The boundaries are interpreted as follows:

- **Video Dwell Time:** Low (0–5 s) represents immediate skip or near-skip behavior, while High (15–30+ s) represents clearly sustained attention to a single item.
- **Negative Sentiment Density:** Low, Medium, and High follow an evenly distributed three-term partition over the 0-100% NSD scale.
- **Session Duration:** Low (0–10 min) represents brief checking, while High (15–40+ min) represents clearly extended sessions.

The rule base is arranged so that increasing Session Duration should not lower risk when the other inputs are fixed, and increasing Negative Sentiment Density should not lower risk. Increasing Video Dwell Time generally raises risk because it reflects more sustained attention to a single item, but that effect is weaker and more context-dependent than the duration and negativity axes. The resulting rule surface is therefore mostly monotonic rather than strictly monotonic.

These ranges are retained as the fixed analytic scoring framework for week-level DSI across both study weeks and as the Week 1 live default. After the 7-day baseline window, if at least 10 sentiment-reliable sessions are available, the live prompt engine updates Session Duration and NSD memberships from that participant's Week 1 $\frac{\frac{Q_{25}}{Q_{50}}}{\frac{Q_{75}}{Q_{95}}}$ quantiles; otherwise the default memberships are retained. The >= 10 threshold is used to avoid unstable personalization from sparse baseline data. Video Dwell Time remains fixed in both analytic and live modes.

Sensitivity Protocol for Fuzzy Boundaries: The study design specified a one-at-a-time local sensitivity protocol for the nine boundary parameters by varying each parameter across a pre-specified plausibility range while the others are held constant. The intended sensitivity outputs are: (1) mean absolute score change, (2) percentage of sessions reclassified (Safe/Warning/Critical), and (3) which parameters produce the largest output instability under the tested range (Dogan, 2021; Shahari & Rasmani, 2024; Shukla & Muhuri, 2025; Vinogradova-Zinkevič, 2023). This protocol is documented for reproducibility and future audit; Chapter 4 does not rely on separate sensitivity-output claims.

- **Rule Evaluation:** The system evaluates a rule base of 27 rules (3 variables × 3 levels each = 3^3 combinations). **Safe** is assigned to clearly low-risk states and isolated mild signals. **Warning** is assigned to mixed or moderate states without strong multi-axis convergence. **Critical** is assigned when prolonged use and negative exposure converge, with Video Dwell Time acting mainly as an intensifier. The rule assignments follow these principles:
  - every Low/Medium/High combination is assigned exactly once, so the engine has no undefined corner cases
  - no rule is **Critical** when Session Duration and Negative Sentiment Density are both Low
  - prolonged duration alone does not produce **Critical** when Negative Sentiment Density remains Low

- High NSD is escalated most strongly when at least one other axis is also non-low, especially Session Duration
- High Dwell Time alone is insufficient for **Critical** and mainly intensifies already concerning states
- **Warning** remains the default mixed class whenever the evidence is concerning but not yet strongly convergent

Read in groups, Rules 1, 2, 4, 10, and 19 are assigned **Safe** because at least two axes remain low. Rules 3, 5-8, 11-14, 16, 20-22, and 25 are assigned **Warning** because they show one elevated axis or mixed evidence without strong duration-negativity convergence. Rules 9, 15, 17-18, 23-24, and 26-27 are assigned **Critical** because prolonged use and negative exposure converge, or because High NSD is reinforced by High Dwell Time in sessions that are no longer brief.

Rule 15 (Medium Dwell, Medium NSD, High Duration) is classified as **Critical** because the session is already prolonged and both other axes are non-low. Rule 23 (High Dwell, Medium NSD, Medium Duration) is also classified as **Critical** because sustained attention and non-low negative exposure already converge in a session that is no longer brief. Rule 25 (High Dwell, High NSD, Low Duration) is classified as **Warning** after psychology expert review because a short session may contain emotionally heavy content without yet showing prolonged or compulsive scrolling. This still flags concentrated negative exposure, but it avoids triggering the strongest intervention from a brief session alone.

Representative rules include:

Table 4

Representative Appendix A Rules (6 of 27)

No.	Video Dwell Time	Negative Sentiment Density	Session Duration	Risk Level
1	Low	Low	Low	Safe
14	Medium	Medium	Medium	Warning
27	High	High	High	Critical
26	High	High	Medium	Critical

No.	Video Dwell Time	Negative Sentiment Density	Session Duration	Risk Level
9	Low	High	High	Critical
19	High	Low	Low	Safe

The complete 27-rule inference base is presented in Appendix A.

Exception Logic: Most rules are arranged so that higher Session Duration or higher Negative Sentiment Density does not reduce risk, but one deliberate exception is retained for interpretive accuracy. Appendix A Rule 9 (Low Dwell, High NSD, High Session Duration) is classified as **Critical** because low dwell should not automatically downgrade a session that is both prolonged and dominated by negative items; in that edge case, rapid chaining across many negative items is treated as more concerning than reassuring. By contrast, Appendix A Rule 19 (High Dwell, Low NSD, Low Session Duration) is not treated as an exception but as a boundary clarification: sustained attention to a single item, without prolonged session time or negative-dominant exposure, is not enough by itself to indicate doomscrolling. This is why the table treats dwell time as a modifier, not the main driver.

- **Defuzzification:** The aggregated fuzzy output is converted back into a crisp "Risk Score" (0–100) using the Center of Gravity (CoG) method. The three output classes occupy equal thirds of the reporting scale, with category cutoffs at 33.33 and 66.67 and output centers at 16.67, 50.00, and 83.33 for Safe, Warning, and Critical, respectively. Formally, this computation involves rule activation weight (w_i) following the MIN T-norm, and output defuzzification:

$$w_i = \min(\mu_{\text{Dwell}}(x), \mu_{\text{NSD}}(y), \mu_{\text{Duration}}(z))$$

$$\text{Risk Score} = \frac{\sum_{i=1}^{n} w_i \cdot c_i}{\sum_{i=1}^{n} w_i}$$

Where w_i is the activation weight of rule i, and c_i is the crisp output center for the mapped risk class. For each fired rule, the smallest membership value determines the rule's activation weight. This prevents one strong input from fully dominating when the other conditions required by that rule remain weak.

Dynamic Weighting vs. Static Percentages: The adopted MIN-based fuzzy inference does not assign flat percentage weights to individual variables. Each fired rule is limited by its weakest required antecedent, so a variable's influence depends on the full combination of inputs rather than on a fixed linear contribution.

Illustrative Worked Example: Given the input tuple (VideoDwellTime = 45 s, NegativeSentimentDensity = 62%, SessionDuration = 35 min), the system computes Video Dwell Time High membership = 1.0, Negative Sentiment Density Medium membership ≈ 0.64, and Session Duration High membership = 0.80. The dominant firing pattern is Appendix A Rule 24 (High Dwell Time, Medium NSD, High Duration → Critical), so the defuzzified Risk Score is 83.33/100 and the resulting intervention level is Level 3 (Critical).

3.6. Method in Evaluation of the Study

The study used four complementary evaluation methods: the **ISO/IEC 25010 Software Quality Model** for technical quality assessment, TAM for user acceptance evaluation, an SME panel for expert review, and a baseline-intervention comparison of selected logged metrics and self-reported doomscrolling scores. Results are reported separately as **operational feasibility** (successful deployment completion and absence of critical system failures), **software acceptability** (ISO/IEC 25010, TAM, and SME results), and **estimator plausibility** (baseline convergent association together with sentiment-reliable coverage).

3.6.1. ISO/IEC 25010 Software Quality Evaluation

The study specifically focuses on four key characteristics of the ISO/IEC 25010 model (International Organization for Standardization, 2023) relevant to privacy-sensitive mobile well-being applications:

1. **Functional Suitability:** Assessing whether the heuristic risk-state estimation and intervention features function as intended and cover the specified requirements.
2. **Performance Efficiency:** Evaluating the application's perceived resource behavior through user survey responses.
3. **Usability:** Measuring the ease of use, learnability, and user satisfaction with the interface and intervention mechanisms.
4. **Reliability:** Assessing the system's stability and availability during extended periods of monitoring without crashing or being killed by the Android OS.

3.6.2. Technology Acceptance Model (TAM) Evaluation

To assess user acceptance, the study uses the Technology Acceptance Model (TAM). Recent mHealth acceptability reviews continue to treat TAM as a practical user-acceptance framework, with Perceived Usefulness and Perceived Ease of Use retained as core constructs (Adnan et al., 2025). Two TAM constructs are measured:

1. **Perceived Usefulness (PU):** The degree to which users believe the system enhances their ability to manage doomscrolling behavior.
2. **Perceived Ease of Use (PEOU):** The degree to which users believe the system is free of effort to learn and operate.

Each construct is measured using 6 Likert-scale items aligned to these two core TAM constructs and adapted to the doomscrolling risk-estimation context. The TAM instrument is administered in the Phase 5 post-usage survey.

3.6.3. Subject Matter Expert (SME) Evaluation

A panel of **two (2) subject matter experts** evaluates the system's technical quality and risk-estimation logic. The SME panel is composed of:

- One (1) expert in software engineering or mobile application development

- One (1) expert in digital well-being, behavioral psychology, or educational technology

The originally planned third reviewer with a dedicated data science, machine learning, or fuzzy-logic specialization could not be engaged within the study period; the rule-base coherence items were therefore reviewed from the software-engineering and behavioral perspectives of the two available experts, and the absence of a dedicated fuzzy-systems reviewer is stated as a limitation of the expert appraisal.

Each SME evaluates the system using a structured rubric covering: (1) the technical soundness of the hybrid risk-estimation framework, including VADER integration and fuzzy inference behavior, (2) the plausibility of the selected input ranges for Video Dwell Time, Negative Sentiment Density, and Session Duration, (3) the internal coherence of the 27-rule base, including whether nearby rules change in a sensible direction across duration and negative exposure and whether exception rules such as rapid negative-content chaining are reasonable, (4) the quality of the system architecture and privacy implementation, (5) the appropriateness of the intervention design, and (6) overall software quality using the same ISO/IEC 25010 characteristics evaluated by end users. SME responses are collected using a 5-point Likert scale and, given the two-member panel, reported descriptively per expert against the study's favorable target rather than as a panel mean with standard deviation, alongside qualitative feedback; the behavioral-psychology expert additionally provided a signed structured narrative validation whose observations are synthesized with the rubric results. This SME review serves as narrative expert appraisal of the rule logic and as an expert plausibility check, not as formal empirical calibration or a content-validity index.

3.6.4. Testing Method

The study uses both **black box** and **white box** testing. Black-box testing checks whether visible functions behave as expected under realistic use, while white-box testing checks whether the internal scoring, routing, and data-flow logic behave correctly at code and module level.

3.6.4.1. Black Box Testing Procedure

The black-box procedure evaluated the application through expected inputs, user actions, and observed outputs without relying on internal code inspection. The full application was tested on a small set of physical devices representing the screen sizes and Android versions available to the researchers to verify that usage patterns (e.g., prolonged session with high negative content) produced the expected risk classification and intervention, including correct routing of no-text cases into the VLM fallback and correct degradation to the 2-input fallback when sentiment remained unresolved. The same procedure checked user-visible flows such as onboarding, permission handling, dashboard display, logging continuity, and prompt presentation.

3.6.4.2. White Box Testing Procedure

The white-box procedure evaluates the internal logic and data flow of the system. The Kotlin-implemented VADER-compatible scoring engine and fuzzy-logic rules are tested in isolation using JUnit to verify that known inputs produce expected outputs. The Accessibility Service-to-database pipeline is also tested to confirm correct data flow and that interventions trigger from logged state changes. This same layer verifies text-path routing, no-text VLM routing, and expected fallback to the 2-input safety mode when sentiment remains unresolved.

3.7. Data Collection Scheduling

Data collection followed five sequential phases. Phase 1 covered recruitment, screening, consent, and group assignment. Phase 2 covered baseline profile completion and application setup. Phase 3 covered the one-week baseline logging period with prompts disabled. Phase 4 covered the one-week deployment period in which only the intervention group received adaptive prompts while the control group continued logging-only. Phase 5 covered the post-usage survey and final retrieval of aggregate local logs.

3.8. Data Gathering

Data gathering was conducted through five phases:

Phase 1: Recruitment, Screening, and Consent Potential participants were recruited through non-probability purposive-convenience sampling using accessible channels such as academic networks, peer referrals, and online groups. Interested individuals underwent screening to confirm eligibility based on age, active use of at least one target short-form video platform, and study-device compatibility, including whether the phone could install the study build, maintain the required permissions, and keep the monitoring service active under ordinary use. Eligible participants were then provided with an informed consent form explaining the study's purpose, procedures, data to be collected, risks, safeguards, and right to withdraw. After consent and eligibility confirmation, participants were assigned to either the intervention group or the logging-only control group using the concealed permuted-block allocation list described earlier, with an initial 1:1 allocation target of 25 per group. Both groups received the same baseline setup and Week 1 logging procedure; the distinction became operational only in Week 2 when prompts were enabled for the intervention group.

Phase 2: Baseline Profile Form and Application Setup Prior to deployment, each participant completed a baseline profile form capturing demographic information, estimated short-form video use, and general platform-use habits. The participant was then assisted in installing the application and enabling the necessary Android permissions. Each participant was assigned a unique study code used to link logs and survey forms; the name-to-code list was retained separately by the researchers. This form provided contextual information but was not treated as the primary source of behavioral measurement.

Week 1 provides the fixed-prior baseline DSI and self-report anchor. At the end of the 7-day baseline, the live prompt engine may lock participant-specific Session Duration and NSD memberships from Week 1 sentiment-reliable sessions while leaving Video Dwell Time fixed. This follows recent JITAI evidence favoring personalized intervention criteria over uniform ones for live prompting (Ikegaya et al., 2025). If reliable baseline coverage is insufficient, the default priors are retained.

Phase 3: Week 1 Baseline Logging Phase During Week 1, the app's built-in logger records quantitative usage metrics locally on the participant's device while adaptive intervention prompts remain inactive. This baseline period establishes the participant's observed usage pattern under the study conditions and provides the baseline DSI and self-report anchor used for convergent association testing.

The Week 1 logged data include:

- **Usage Metrics:** Session duration, video dwell time, and supporting interaction counts such as swipe totals.
- **Sentiment-Related Indicators:** Session-level sentiment scores and Negative Sentiment Density estimates.
- **Reliability Events:** Application crashes, service interruptions, and related system-stability notes.

At the conclusion of Week 1, participants completed the short-form 4-item **Doomscrolling Scale** (Satici et al., 2023; Sharma et al., 2022), using Items 1, 2, 10, and 12 of the original instrument and a 7-day recall instruction stem (e.g., "In the past week…"; see Appendix E), to provide the baseline self-report anchor for convergent association. The original instrument was validated as a general self-report scale, so changes between administrations are interpreted as short-horizon self-report differences.

Phase 4: Week 2 Deployment Phase During Week 2, **intervention-group** participants received adaptive digital mindfulness prompts through a live prompt engine that retained the fixed rule base and fixed Video Dwell Time memberships but, when available, replaced the global Session Duration and NSD memberships with participant-specific Week 1 quantile-derived bounds. **Control-group** participants continued logging-only operation with prompts disabled via a configuration flag. Both groups continued to have usage and sentiment-related metrics logged, while prompt-response data for the intervention group were recorded descriptively.

For behavioral comparison, raw elapsed session duration and raw elapsed video dwell time were treated as the primary time-based metrics. **Prompt-excluded active-use metrics** were also retained: when a prompt appeared, the active content-view interval was closed, and time spent inside the system-generated intervention was excluded until the participant returned to the target platform. These prompt-excluded metrics are reported as supplementary traces of prompt-interrupted use. Observed Week 2 differences are interpreted as short-term behavioral differences under the study conditions.

Potential confounds include time-of-day and day-of-week variation, external events during the two-week window, and device-specific differences in processing speed or background-process management. The randomized two-group design is intended to help balance these factors across study arms, while the within-subject Week 1 to Week 2 comparison in the intervention group accounts for stable individual differences.

At the conclusion of Week 2, participants completed the same short-form 4-item **Doomscrolling Scale** under the same 7-day recall instruction stem (Appendix E) to provide the post-intervention self-report anchor for paired comparison.

Phase 5: Post-Usage Survey and Final Data Retrieval To align data structures for baseline convergent association and descriptive week-level comparison, logged session summaries over each 7-day period are aggregated into a composite **Doomscroll Severity Index** per user per week. This analytic DSI uses the same fixed-prior scoring framework across both study weeks. Unlike the live Week 2 prompt engine, the DSI engine is not personalized after Week 1, so week-level summaries remain on one fixed scale:

$$\mathrm{DSI}_w = \left(\frac{1}{m_w}\right)\sum_{s=1}^{m_w} \mathrm{RiskScore}_s$$

where m_w is the number of week-w sessions classified as sentiment-reliable and therefore doomscrolling-evaluable. Sessions flagged as **Sentiment-Unreliable** are excluded from DSI computation. DSI is a mean of retained session scores rather than a cumulative sum, so these exclusions affect coverage more than scale. Week-specific DSI values are retained for descriptive comparison, convergent testing is anchored to the baseline week only, and DSI coverage is reported by week and study arm in terms of retained participants, retained sessions, and excluded sentiment-unreliable sessions.

Interpretation of Classification Metrics: Traditional supervised classification metrics (e.g., precision, recall, F1, AUROC) are not estimated because session-level ground-truth labeling would require interrupting active scrolling. The study therefore uses convergent association as the main plausibility check and interprets the system as a heuristic behavioral risk estimator.

After the deployment period, participants completed a structured post-usage survey instrument with three components: (a) the ISO/IEC 25010 software quality evaluation, (b) the TAM assessment, and (c) open-ended feedback items. At the same stage, aggregate local logs were exported using the participant study code only and transferred to a password-protected research folder accessible only to the researchers.

For the ISO/IEC 25010 component, the survey uses a 5-point Likert Scale to evaluate the four target quality characteristics: Functional Suitability, Performance Efficiency, Usability, and Reliability. For the Usability sub-characteristic, all 10 items of the System Usability Scale (SUS) are adopted verbatim, and the standard SUS scoring formula is applied to yield a composite usability score on a 0–100 scale. The SUS is a standardized usability measurement tool whose validity for digital health applications has been supported by a meta-analysis of 114 apps yielding a benchmark mean of 68.05 (SD 14.05) (Hyzy et al., 2022). For the remaining three sub-characteristics (Functional Suitability, Performance Efficiency, and Reliability), the study uses researcher-developed items derived from the ISO/IEC 25010 sub-characteristic definitions (International Organization for Standardization, 2023) and refined through pre-deployment expert review.

For the TAM component, 12 Likert-scale items (6 for Perceived Usefulness, 6 for Perceived Ease of Use) aligned to the study's contextualized TAM constructs are included to assess user acceptance of the system.

3.8.1. Instrument Review and Reliability Safeguards

Before deployment, the combined survey instrument underwent structured expert content review to check item relevance, clarity, redundancy, and alignment with the study constructs. Reviewer comments were consolidated, and unclear or overlapping items were revised before administration.

After data collection, internal consistency for each multi-item sub-scale was estimated through Cronbach's alpha and interpreted as an index of item coherence, with appropriate caution given the pilot sample size and the mix of standardized and adapted items.

Open-ended feedback items were used as descriptive support for the survey findings. Two researchers independently reviewed the responses, reconciled coding differences through discussion, and reported recurring issue categories.

The alignment between the study's research variables, data sources, and instruments is summarized in the following table.

Table 5

Variable–Instrument Alignment Matrix

Research Variable	Data Source	Instrument / Tool	Phase
Baseline usage profile	Self-report / Logs	Baseline profile form + Week 1 logs	Phase 2 & 3
Session Duration	Automated system logs	Built-in application logger	Phase 3 & 4
Video Dwell Time (s)	Automated system logs	Built-in application logger	Phase 3 & 4
Negative Sentiment Density (%)	Automated system logs	VADER text-path + VLM no-text path + Built-in logger	Phase 3 & 4
Prompt-response profile (descriptive)	Automated system logs	Built-in application logger	Phase 4
Pre/Post Doomscrolling Scale Score	Self-report	Short-form 4-item Doomscrolling Scale (Appendix E)	End of Phase 3 & 4
Baseline convergent association	Week 1 DSI + Week 1 Doomscrolling Scale	Bivariate Correlation (Pearson's r / Spearman's rho)	End of Phase 3
Functional Suitability	Likert-scale survey	Researcher-developed ISO/IEC 25010 items + expert review	Phase 5
Performance Efficiency (subjective)	Likert-scale survey	Researcher-developed ISO/IEC 25010 items + expert review	Phase 5

Table 5. Variable–Instrument Alignment Matrix (continued)

Research Variable	Data Source	Instrument / Tool	Phase
Usability	Likert-scale survey	System Usability Scale (SUS), benchmarked in Hyzy et al. (2022)	Phase 5
Reliability	Likert-scale survey	Researcher-developed ISO/IEC 25010 items + expert review	Phase 5
Perceived Usefulness (TAM)	Likert-scale survey	Contextualized TAM PU items	Phase 5
Perceived Ease of Use (TAM)	Likert-scale survey	Contextualized TAM PEOU items	Phase 5
Expert quality assessment	SME evaluation	Structured rubric (ISO 25010 + risk-estimation logic)	SME

3.9. Respondents of the Study and Sampling Technique

The respondents of this study were Filipino Android users aged 18 years and above who actively used at least one of the target short-form video platforms. Findings apply to this target population and to the non-probability nature of the sample. The sampling technique employed was **purposive-convenience sampling** because participants had to meet specific technical and behavioral eligibility criteria and be reachable through channels accessible to the researchers.

The study reports how many participants used TikTok, Facebook Reels, and Instagram Reels during the field evaluation, including platform mix by study arm. Per-platform extraction success rates and analyzable-session counts are also reported, and cross-platform interpretation is limited to platforms that yielded stable extraction throughout the deployment window.

Inclusion Criteria:

- Must be aged 18 years or older to provide independent informed consent.
- Must own an Android smartphone (Version 8.0 or higher) that passes study-compatibility screening for installation, permissions, and background-service stability.
- Must be an active user of at least one target short-form video platform (TikTok, Facebook Reels, or Instagram Reels).
- Must consent to the installation of the application and the collection of usage data.

Sample Size Justification: A minimum of **N = 50** participants is targeted, split as evenly as possible into an intervention group and a control group, with an initial allocation target of **n = 25** per group. This target supports a balanced pilot deployment and post-usage evaluation within an undergraduate thesis scope.

Analyses that exclude sentiment-unreliable sessions or participants with insufficient baseline data report their effective sample sizes separately. With a minimum sample of about 50 participants, between-group estimates are interpreted with emphasis on effect sizes and 95% confidence intervals alongside p-values.

Selection Bias Acknowledgment: Purposive-convenience sampling may overrepresent users who are already receptive to self-regulation tools. This self-selection tendency can affect perceived acceptance and compliance estimates.

3.10. Statistical Treatment

Research Question 1 is addressed descriptively through the design-and-development sections earlier in this chapter. The evaluation data are analyzed using the following statistical methods, each selected for a specific data source and objective.

1. Descriptive Summaries (Survey Data — ISO 25010 & TAM): The Likert-scale responses from the ISO/IEC 25010 software quality evaluation survey and the TAM instrument (Perceived Usefulness and Perceived Ease of Use subscales) are summarized using **mean scores** and standard deviation.

Mean scores are described on a 5-point Likert scale, where 3.00 represents the neutral midpoint. For evaluative decisions, the study uses pre-set favorable thresholds of >= 3.50 for ISO/IEC 25010 and TAM constructs and >= 4.00 for SME ratings. These are study-defined favorable benchmarks rather than universal cutoffs.

2. Objective Acceptance Thresholds for Evaluation: To reduce post-hoc rationalization of evaluation results, the following pre-set interpretation thresholds are established as decision aids for Chapter 4 rather than as stand-alone proof of software success:

- **System Usability Scale (SUS):** A composite score of >= 70 is treated as the study's usability target, set slightly above the 68.05 digital-health benchmark mean reported by Hyzy et al. (2022).
- **ISO/IEC 25010 Subscales:** A mean score of >= 3.50 out of 5 is treated as the study-defined favorable-evaluation target for Functional Suitability, Performance Efficiency, and Reliability.
- **TAM Constructs:** A mean score of >= 3.50 is treated as the study-defined favorable-evaluation target for Perceived Usefulness and Perceived Ease of Use.

- **SME Evaluation:** A rating of >= 4.00 out of 5 is treated as the study-defined favorable-evaluation target for expert review; given the two-member panel, rubric ratings are reported per expert against this target.

Results below these thresholds are reported as unmet software-acceptability targets in Chapter 4. Operational feasibility and estimator plausibility are reported separately. Given the pilot-sized sample and the possibility of analyzable-sample reduction from sentiment-unreliable sessions, inferential results are interpreted primarily through effect sizes, confidence intervals, and coverage rather than pass-fail significance alone.

3. Cronbach's Alpha (Instrument Reliability): To assess internal consistency, **Cronbach's alpha** (α) is computed for each sub-scale (Functional Suitability, Performance Efficiency, Usability, Reliability, PU, PEOU). A pre-set coefficient threshold of $\alpha \geq 0.70$ is adopted as a conventional descriptive benchmark for this study. Given the pilot sample and the mix of standardized and adapted items, alpha is treated as a reliability check rather than as a stand-alone validation claim.

4. Primary Behavioral Analysis: The main quantitative comparison for Research Question 2 uses an **Independent Samples t-test on change scores** (Week 2 minus Week 1) to compare observed short-term behavioral differences between the intervention group and the control group. Using change scores rather than raw Week 2 values adjusts for chance baseline imbalance and individual differences. As a secondary within-subject analysis, a **Paired Samples t-test** compares Week 1 and Week 2 within the intervention group. Given the pilot sample, these tests are treated as exploratory analyses of short-term observed differences under the study conditions. Separate primary tests are conducted for:

- Mean daily **raw elapsed** session duration (minutes)
- Mean daily **raw elapsed** video dwell time (s)
- Mean daily Negative Sentiment Density (%)

- Pre/Post Doomscrolling Scale Score (Self-Reported)

For the Negative Sentiment Density outcome, mean daily NSD is computed only from sentiment-reliable sessions. Days or participant-weeks with no sentiment-reliable sessions contribute missing data for that specific outcome and are excluded only from that specific NSD comparison, with the analyzed n reported.

As a supplementary transparency check, the same tests are also conducted on **prompt-excluded active-use metrics** for session duration and video dwell time. The difference between raw elapsed and prompt-excluded results is reported to show the measurement-rule effect.

To control family-wise Type I error across the four change-score comparisons, p-values were adjusted using the **Holm-Bonferroni sequential correction** at an initial family-wise $\alpha = 0.05$. For each comparison, **Cohen's d** and its 95% confidence interval were reported alongside the Holm-adjusted p-value to support interpretation of effect size, not statistical significance alone.

Handling of Assumption Violations: For the primary independent-samples t-test, normality of change scores within each group is checked with Shapiro-Wilk and equality of variances with Levene's test. If normality is severely violated, the **Mann-Whitney U test** is used as fallback. If variances are unequal but normality holds, Welch's t-test is applied. For the secondary paired t-test, normality of within-subject differences is checked with Shapiro-Wilk; if violated, the **Wilcoxon Signed-Rank Test** is used. If analyzable sample sizes become too small for stable inferential interpretation after outcome-specific exclusions, the affected comparison is reported descriptively with its effective n and directional pattern rather than being overstated as confirmatory evidence.

5. Descriptive Analysis of Prompt-Response Logs: Week 2 prompt-response data (e.g., continue, dismiss, take-break) are summarized using frequency counts and percentages. These logs are treated as descriptive evidence of intervention interaction only and are not analyzed as a baseline-comparable outcome variable.

6. Baseline Convergent Association Analysis: Baseline convergent association is assessed on **Week 1 data only**. First, a component-wise correlation table reports the bivariate association between each individual input variable (mean daily session duration, mean daily video dwell time, mean daily NSD) and the baseline Doomscrolling Scale score. Second, the bivariate correlation between the full composite DSI (computed with fixed priors) and the Doomscrolling Scale is computed. The pattern of component-wise versus composite correlations is discussed descriptively to assess whether the multi-variable composite associates more strongly with self-reported doomscrolling than session duration alone.

Only participants with at least three sentiment-reliable sessions in Week 1 are included; the effective sample size ($N_{\text{effective}}$) is reported. This coverage rule follows recent digital-measure practice for forming week-level aggregates from repeated observations, with three observations commonly used as a pragmatic weekly minimum while stronger reliability claims require more data (Buekers et al., 2025; Meyer et al., 2022; Ratitch et al., 2023; Yao et al., 2021). Here, the >= 3 rule functions as a pragmatic floor for the baseline aggregate. **Pearson's r** is used when linearity and normality assumptions hold; otherwise **Spearman's ρ**. Results are interpreted as a baseline plausibility check rather than as diagnostic validation.

7. Interpretation: For these exploratory pilot analyses, the nominal level of significance is set at $\alpha = 0.05$. Statistically significant differences between Week 1 and Week 2 are interpreted as observed short-term behavioral differences under the study conditions. The two-group randomized design supports stronger inference than a one-group pre/post comparison, but findings are not interpreted as long-term intervention effects or calibrated validation. Prompt-excluded active-use metrics are interpreted separately because they incorporate the prompt-display rule.

8. Descriptive Coding of Open-Ended Feedback (Phase 5): Open-ended responses from the post-usage survey were analyzed through a light inductive coding procedure. Two researchers read the responses in full, grouped them into recurring issue categories, and reconciled coding differences through discussion before the categories were summarized. The goal was to identify usability issues, perceived intervention value, and implementation constraints not captured by the Likert-scale measures. This step supports interpretation of the survey results and surfaces recurring concerns for Chapter 4 discussion.

9. Missing Data and Exclusion Criteria: For the convergent correlation analysis, participants need at least three sentiment-reliable sessions in Week 1 ($m_1 \geq 3$) for Week 1 DSI to enter the correlation dataset. This rule includes the edge case $m_1 = 0$, where DSI is undefined by construction. The effective sample size ($N_{\text{effective}}$) after this exclusion is reported explicitly. For the NSD outcome, participants require at least one sentiment-reliable day in each compared week to contribute a valid week-level comparison; otherwise that specific outcome is treated as missing for that participant. For paired pre/post analyses, complete paired observations are required per outcome variable, and the analyzed n for each test is reported.

3.11. Ethical Considerations

This study adheres to ethical standards in participant data collection and usage. Participants are informed of the study's purpose, the data to be collected, and their right to participate voluntarily or withdraw from the study. The study version also requires explicit user activation of the Android Accessibility Service through the device's system settings, providing a clear operational consent step for text extraction, interaction monitoring, and the service-declared screenshot capability used for no-text visual processing.

Personal identifiers are not included in the research dataset. Logs and survey responses are linked only through participant study codes, while any name-to-code reference list is kept separately by the researchers. Consistent with the Data Privacy Act of 2012, collected data are handled, stored, and protected only for academic research purposes. The system processes scraped text content and temporary no-text screen frames in RAM only, does not retain raw text or captured frames, and stores only aggregate metrics such as sentiment scores, session duration, risk scores, and prompt-response summaries.

The system does not transmit routine study data to external servers and does not store sensitive personal information beyond what is necessary for analysis. Exported aggregate logs are kept in a password-protected research folder accessible only to the researchers. Participants may withdraw from the study at any time, and locally stored usage data linked to their study code may be deleted upon withdrawal.

The study does not involve clinical diagnosis or treatment of mental health conditions. The pause-and-reset prompts, break reminders, and short breathing breaks are designed as general digital wellness prompts, not therapeutic interventions.

The methodological and technical limitations arising from the research design and system architecture of this study are documented in Chapter 1 under **Scope, Delimitations, and Limitations**.

Chapter 4

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the completed field-study, survey, expert-review, and workbook-based analysis results for the Heuristic Risk-State Estimation System. The final analysis dataset contains 50 participants, divided evenly between the intervention group and the logging-only control group. Both groups completed the Week 1 baseline phase and the Week 2 deployment phase. The analysis is interpreted as a pilot-scale software-engineering evaluation, not as a clinical trial or long-term behavioral-efficacy study. To keep results traceable to Chapter 1, this chapter is organized around the five Statements of the Problem (SOPs): each top-level section reports the evidence, findings, and interpretation for one SOP.

4.1. Presentation of Results

This section presents the preliminary context needed to interpret the SOP-level results that follow: the analysis workbook that consolidates the reported statistics, the data sources it draws from, the effective sample sizes after data-quality screening, and the respondent baseline profile. To keep the discussion aligned with Chapter 3, results are organized by SOP and interpreted with attention to effective sample sizes, pilot-study scope, and known data limitations.

4.1.1. The REINVENT Analysis Workbook

The detailed workbook sheets in `REINVENT ANALYSIS.xlsx` were used as the primary source for the tables in this chapter. The workbook is the consolidated analysis artifact for the study: it ingests the raw exported CSV logs from the deployed Android application (session, prompt, reliability, and survey files) and derives every reported statistic through documented cell formulas, so that any reader can trace a Chapter 4 number back to the sheet, column, and formula that produced it. The workbook is organized into four sheet families: (1) documentation and audit sheets that record the analysis intent, the statistician sign-off, and the mapping between reported values and the reviewed manuscript; (2) derived analysis sheets grouped by SOP; (3) supporting derivation and formula-audit sheets; and (4) the raw exported CSVs kept as embedded reference tabs so that the workbook remains self-contained.

The final reporting values for SOP 2 were taken from the sheet `RQ2_FINAL_TESTS_PDF_ALIGNED` because it applies the methodology-selected tests after assumption checking; earlier `RQ2_CHANGE_TESTS` and `RQ2_PAIRED_INTERVENTION` sheets are retained only as intermediate calculation references. The `STATISTICIAN_SIGNOFF` and `PDF_ALIGNMENT_NOTES` sheets record the review of the numbers against the manuscript, and the `FORMULA_NOTES` sheet documents the derivations used in the derived sheets. The formula audit sheets were used only as supporting calculation references.

Table 6

Sheet Families in `REINVENT ANALYSIS.xlsx`

Family	Representative Sheets	Purpose
Documentation and audit	`README`; `GUIDE`; `QUESTIONS`; `STATISTICIAN_SIGNOFF`; `PDF_ALIGNMENT_NOTES`; `EXECUTIVE_SUMMARY`; `FORMULA_NOTES`	Describe workbook usage, record analysis decisions, and align workbook values to the reported manuscript
SOP 1 - architecture and data quality	`DATA_QUALITY_SUMMARY`; `PROMPT_SUMMARY`; `mvl_concordance`	Summarize session reliability, prompt-response counts, and Filipino MVL concordance
SOP 2 - behavioral change	`WEEK_METRICS`; `CHANGE_SCORES`; `RQ2_CHANGE_TESTS`; `RQ2_PAIRED_INTERVENTION`; `RQ2_FINAL_TESTS_PDF_ALIGNED`; `WEEK_DAILY`; `RQ2_GROUP_DATA`	Derive Week 1 and Week 2 aggregates, change scores, and the assumption-selected final tests reported for SOP 2

Family	Representative Sheets	Purpose
SOP 3 - baseline correlation	RQ3_CORRELATION; RQ3_DATA	Compute Spearman's rho and Pearson's r between Week 1 DSI components and the Doomscrolling Scale
SOP 4 - user evaluation	SURVEY_ANALYSIS; ISOIEC 25010; Usability(SUS); TAM(PUPEOU); SURVEY_SCORES	Derive per-construct means, standard deviations, thresholds, and Cronbach's alpha for ISO/IEC 25010, SUS, and TAM
Raw exported CSVs	sessions.csv; daily_summaries.csv; study_periods.csv; prompt_events.csv; reliability_events.csv; risk_personalization.csv; doomscrolling_scale.csv; survey_iso25010.csv; sus_responses.csv; tam_responses.csv; open_ended_feedback.csv; sme_evaluation.csv; sme_open_ended_feedback.csv; baseline_profile.csv	Preserve the source data exported by the Android application and the survey/SME instruments; every derived sheet references these tabs so the workbook is self-contained

The Data Sources and Analysis Mapping table below identifies the specific workbook sheet or exported CSV used to report each SOP.

Table 7

Data Sources and Analysis Mapping

Research Area	Primary Workbook Source	Analysis or Summary Used	Reporting Purpose
Respondent profile	baseline_profile.csv	Counts and descriptive summaries	Describe participant context
Session coverage	sessions.csv; DATA_QUALITY_SUMMARY	Reliable and sentiment-unreliable session counts	Report analyzable coverage and exclusions
SOP 2 behavioral comparison	RQ2_FINAL_TESTS_PDF_ALIGNED	Assumption-selected between-group and paired tests	Report Week 1-to-Week 2 differences
SOP 3 association	RQ3_CORRELATION	Spearman's rho and Pearson's r	Assess baseline convergent plausibility
SOP 4 user evaluation	SURVEY_ANALYSIS	Means, SDs, thresholds, Cronbach's alpha	Evaluate ISO/IEC 25010, SUS, and TAM
SOP 5 expert evaluation	sme_evaluation.csv; sme_open_ended_feedback.csv	Per-expert rubric ratings and narrative comments	Report expert plausibility appraisal

4.1.2. Sample Disposition and Effective Sample Sizes

All 50 allocated participants completed both study weeks. The intervention and control groups each contained 25 participants. The full session log contained 10,134 sessions across the two-week period. Of these, 8,817 sessions were sentiment-reliable and 1,317 sessions were sentiment-unreliable. The sentiment-unreliable sessions were retained as operational evidence but excluded from DSI and NSD computations where the methodology required reliable sentiment coverage.

Table 8

Sample Disposition and Effective Sample Sizes

Metric	Intervention	Control	Total	Interpretation
Allocated participants	25	25	50	Balanced two-arm pilot sample
Completed Week 1	25	25	50	All participants retained
Completed Week 2	25	25	50	All participants retained
SOP 3 eligible participants	23	25	48	Two intervention participants lacked sufficient reliable Week 1 sessions
Primary duration, dwell, and self-report outcomes	25	25	50	Complete outcome pairs available
NSD and DSI outcomes	23	25	48	Effective n reflects sentiment-reliability coverage

The final session distribution was 4,820 intervention-group sessions and 5,314 control-group sessions. By platform, the logs contained 3,433 Instagram sessions, 3,389 Facebook sessions, and 3,312 TikTok sessions. Week 1 contained 5,162 sessions, while Week 2 contained 4,972 sessions. This distribution shows that the application captured data from all target platforms and both study phases.

4.1.3. Respondent Baseline Profile

The respondent profile provides descriptive context for the behavioral analyses. The baseline profile sheet shows that the participants were adult Android users aged 18 to 29 years, with a mean age of 22.12 years. There were 39 male and 11 female participants. All

50 participants had observed use of Facebook Reels and Instagram Reels during Week 1, while 49 had observed TikTok use. Estimated daily short-form-video use ranged from 15 minutes to 883 minutes, with a mean of 392.70 minutes based on the active-day derivation used in the completed baseline profile sheet.

Table 9

Respondent Baseline Profile Summary

Profile Variable	Result	Interpretation
Total respondents	50	Balanced sample with 25 intervention and 25 control participants
Age	18-29 years; mean 22.12	Adult participant group, mostly young adults
Sex	39 male; 11 female	Male respondents formed the larger share of the sample
Observed Facebook Reels use	50 of 50	All participants had Facebook Reels activity
Observed Instagram Reels use	50 of 50	All participants had Instagram Reels activity
Observed TikTok use	49 of 50	Nearly all participants had TikTok activity
Estimated daily SFV use	15-883 min; mean 392.70	Wide variation in baseline short-form-video exposure

In terms of general platform-use habits, 25 participants were categorized as having multiple sessions daily, 15 as having many sessions through the day, and 10 as having a few sessions daily. Dominant-platform descriptors were distributed across Facebook Reels, Instagram Reels, and TikTok. These profile results support the suitability of the sample for a pilot evaluation of a short-form-video monitoring application, while the purposive-convenience sampling still limits generalizability.

4.2. SOP 1: Privacy-Preserving Mobile Architecture and Estimation Framework with 2-Input Behavioral Fallback for Sentiment-Unreliable Sessions

Statement of the Problem 1 asked what privacy-preserving mobile architecture and estimation framework can support doomscrolling-related risk estimation on the target short-form video platforms using behavioral indicators and sentiment-related indicators when reliably resolvable, with 2-input behavioral fallback for sentiment-unreliable sessions. This section reports the implemented architecture, its operational data-quality evidence, and the analytic time complexity of the runtime pipeline.

4.2.1. System Implementation and Data-Quality Evidence

The completed system implements the architecture described in Chapter 3: Android Accessibility Service monitoring, local session tracking, text-first VADER-compatible sentiment scoring, Filipino/Taglish MVL extension, no-text visual fallback through transient on-device screenshot processing, fuzzy inference, adaptive prompting, local Room storage, and CSV export using participant study codes.

The implementation evidence supports the intended privacy boundary. Raw text and no-text screen frames are processed locally and are not retained as research outputs. The exported study data contain aggregate behavioral and sentiment-related metrics rather than raw captions, comments, or images.

The Filipino MVL concordance sheet contained 61 rows, including the 57 runtime lexicon entries (56 affective entries and one neutral candidate entry) with exact expert-to-runtime valence agreement. All 57 runtime entries were within the acceptable plus-or-minus-one valence window. This supports the internal consistency of the deployed Filipino/Taglish lexicon values against the completed single-expert review. Because the review used one Filipino language expert, this result is treated as a concordance check, not as inter-rater reliability.

The deployment also logged 1,329 reliability events: 1,317 content-related events (high-OOV, VLM-unresolved, and extraction-failure), each linked to one sentiment-unreliable session, plus 12 service start/stop notices. The event percentages below use this 1,329-event total as the denominator.

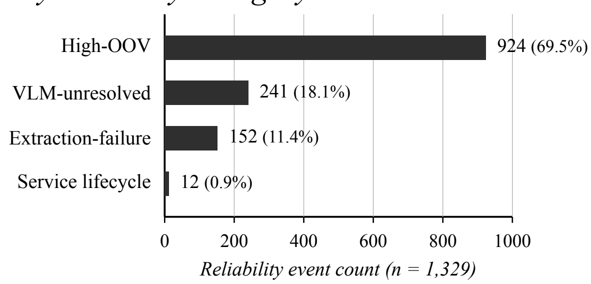
Table 10

Data Quality and Session Reliability Summary

Category	Count	Percent	Interpretation
Reliable sessions	8,817	87.0%	Included in sentiment-dependent analysis
Sentiment-unreliable sessions	1,317	13.0%	Excluded from DSI or NSD where required
High-OOV reliability events	924	69.5% of reliability events	Main reliability-event driver
VLM-unresolved reliability events	241	18.1% of reliability events	No-text visual path did not resolve some cases
Extraction-failure events	152	11.4% of reliability events	Platform or accessibility extraction issue
Service lifecycle events	12	0.9% of reliability events	Service start/stop notices not tied to sessions
Total logged sessions	10,134	100.0%	Operational data captured across both weeks

Figure 4

Distribution of Reliability Events by Category



The bar chart makes visible what the reliability table only implies: high-OOV text was the dominant driver of sentiment-unreliability, accounting for roughly seven out of every ten reliability events. Screenshot-plus-VLM handling and platform-side extraction failures shared the remaining content-related events, and service-lifecycle notices contributed less than one percent.

Prompt logs were available for the Week 2 intervention arm. The largest prompt-related action was `TAKE_BREAK`, with 191 logged events or 51.6% of prompt events. `SHOWN` accounted for 68 events, `SUPPRESSED` for 49, `CONTINUE` for 45, `DISMISSED` for 17, and `VIEW_DASHBOARD` for 0. These logs are descriptive evidence of intervention interaction. They are not interpreted as a separate behavioral outcome because there is no Week 1 prompt baseline.

Table 11

Prompt-Response Summary During Week 2 Intervention

Prompt Action	Count	Percent	Interpretation
`TAKE_BREAK`	191	51.6%	Most frequent logged prompt response
`SHOWN`	68	18.4%	Prompt display engine state
`SUPPRESSED`	49	13.2%	Cooldown or suppression state
`CONTINUE`	45	12.2%	User continued after prompt
`DISMISSED`	17	4.6%	User dismissed the prompt
`VIEW_DASHBOARD`	0	0.0%	No dashboard-view event was logged from prompts

These findings answer SOP 1 by showing that the implemented system can collect privacy-bounded local logs, estimate session-level risk from observable proxies, degrade to conservative fallback handling when sentiment is unreliable, and record intervention events for descriptive analysis.

4.2.2. Algorithm Complexity

This subsection presents the time complexity of the core runtime algorithms described in Chapter 3, closing out SOP 1 by characterizing the computational cost of the implemented architecture. The analysis is analytic and data-independent: it characterizes the computational cost of the implemented pipeline as a function of its input sizes and does not depend on field-study data. The following variables are used:

- n - number of tokens in the text payload extracted from a single viewed item
- I - number of items viewed in a session
- V - number of fuzzy input variables ($V = 3$; $V = 2$ in the degraded fallback mode)
- L - number of linguistic terms per variable ($L = 3$)
- R - number of rules in the fuzzy inference base ($R = L^V = 27$; $R = 9$ in fallback mode)

- S - number of recorded sessions in a study week
- C_{VLM} - the bounded per-invocation cost of one Moondream 0.5B inference

VADER Sentiment Scoring. Tokenization visits each token once. Lexicon membership is resolved through hash-based dictionary lookup with constant average cost per token, and the Filipino MVL extension only enlarges the dictionary without changing the lookup mechanism. The booster and negation heuristics inspect a fixed lookback window of at most three preceding tokens per scored token, which is a bounded constant amount of work. Per-item text scoring is therefore $O(n)$ time, and the compound-score normalization $C = x/\sqrt{x^2 + \alpha}$ is $O(1)$.

Per-Item Routing Heuristic. Deciding whether a viewed item carries usable text is a constant-time check, so routing contributes $O(1)$ per item.

No-Text VLM Fallback (Moondream 0.5B). Each no-text item triggers at most one screenshot capture and one constrained VQA inference. Because the model size, the input resolution, and the constrained five-label output are all fixed, every invocation has a bounded constant cost C_{VLM} that does not grow with session length. C_{VLM} is the dominant constant in the pipeline, which is why the routing heuristic reserves it for items with no usable text instead of applying it universally.

Fuzzy Inference Engine. One session-level inference consists of: (1) fuzzification, which evaluates $V \cdot L = 9$ triangular membership functions at $O(1)$ each; (2) rule evaluation, which computes a MIN over V antecedent memberships for each of the R rules, i.e., $O(R \cdot V)$; and (3) CoG defuzzification, a weighted average over at most R activation weights, i.e., $O(R)$. Since V, L, and R are fixed at runtime, a complete inference runs in constant time, $O(R \cdot V) = O(1)$, in both modes.

Session Aggregation and Week-Level DSI. Negative Sentiment Density is a ratio over resolved items maintained incrementally in $O(I)$ per session. Week-level DSI computation aggregates session records in $O(S)$. Week 1 personalization derives the

$Q_{25}/Q_{50}/Q_{75}/Q_{95}$ duration and NSD quantiles from at least 10 sentiment-reliable baseline sessions, which is $O(S \log S)$ due to sorting.

Table 12

Time Complexity Summary of Core Runtime Algorithms

Component	Time Complexity	Dominant Cost Driver
VADER scoring (per item)	$O(n)$	Token count of extracted text
Per-item routing heuristic	$O(1)$	Constant-time payload check
VLM fallback (per no-text item)	$O(C_{\text{VLM}})$	Fixed-size Moondream 0.5B inference
Fuzzy inference (per session)	$O(R \cdot V) = O(1)$	Fixed 27-rule or 9-rule base
NSD aggregation (per session)	$O(I)$	Items viewed in the session
Week-level DSI	$O(S)$	Sessions recorded in the week
Week 1 quantile personalization	$O(S \log S)$	Sorting baseline sessions

Worst-Case Analysis. For a single session, the total cost is $O\left(\sum_{i=1}^{I} n_i\right)$ for text-path items plus $O(I \cdot C_{\text{VLM}})$ for VLM-routed items, followed by one $O(1)$ fuzzy inference. No component is super-linear in the number of viewed items or tokens. The worst case occurs in a feed where every viewed item lacks usable text, so every item incurs the screenshot-plus-VLM cost, giving $O(I \cdot C_{\text{VLM}})$ for the session. If screenshot capture or VLM inference fails, the safety lock degrades inference to the 2-input, 9-rule engine without adding asymptotic cost, and the session is marked Sentiment-Unreliable as described in Chapter 3.

4.3. SOP 2: Short-Term Week 1-to-Week 2 Changes in Logged Usage Metrics and Self-Reported Doomscrolling Scores

Statement of the Problem 2 asked what short-term Week 1-to-Week 2 changes are observed in selected logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, and within the intervention group across the same period. The final reporting table uses assumption-selected tests: Welch's t-test for session duration where normality held with unequal variance, Mann-Whitney U tests for non-normal between-group change scores, paired t-tests for normally distributed within-intervention differences, and Wilcoxon signed-rank tests for non-normal paired differences.

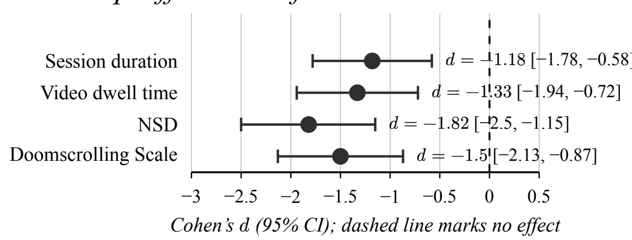
Table 13

SOP 2 Primary Behavioral Comparisons Between Groups

Outcome	n (I/C)	Final Test	Statistic	p / Holm	Effect Size	95% CI
Session duration (min)	25 / 25	Welch t-test	$t = -4.17$	$p < .001$	$d = -1.18$	$[-1.78, -0.58]$
Video dwell time (s)	25 / 25	Mann-Whitney U	$U = 124.0$	$p < .001$	$d = -1.33$	$[-1.94, -0.72]$
NSD (%)	23 / 25	Mann-Whitney U	$U = 68.0$	$p < .001$	$d = -1.82$	$[-2.50, -1.15]$
Doomscrolling Scale	25 / 25	Mann-Whitney U	$U = 105.0$	$p < .001$	$d = -1.50$	$[-2.13, -0.87]$

Figure 5

Forest Plot of Between-Group Effect Sizes for the Four SOP 2 Primary Outcomes



Every 95% confidence interval sits fully to the left of the zero-effect reference line, corresponding to reductions in the intervention arm for each outcome. NSD shows the largest magnitude and the interval farthest from zero, while session duration has the smallest magnitude of the four; the intervals overlap enough that none of the four outcomes can be singled out as decisively larger than the others.

All four primary outcomes favored the intervention group after Holm-Bonferroni correction. The negative effect sizes indicate larger reductions in the intervention group compared with the logging-only control group. Session duration, dwell time, NSD, and self-reported Doomscrolling Scale scores all showed short-term reductions under the intervention condition. Because the study is pilot-sized and lasted two weeks, these results are interpreted as observed short-term differences under study conditions, not as proof of long-term intervention efficacy.

Table 14

SOP 2 Supplementary and Within-Intervention Comparisons

Outcome	n	Final Test	Statistic	p	Effect Size	Interpretation
Mean sessions/day (between groups)	25 / 25	Mann-Whitney U	$U = 153.5$	$p = .002$	$d = -0.82$	Supplementary; favors intervention
DSI composite risk (between groups)	23 / 25	Mann-Whitney U	$U = 81.0$	$p < .001$	$d = -1.84$	Supplementary; favors intervention
Session duration (within intervention)	25	Paired t-test	$t = -4.41$	$p < .001$	$d_z = -0.88$	Reduction from Week 1 to Week 2
Video dwell time (within intervention)	25	Paired t-test	$t = -5.45$	$p < .001$	$d_z = -1.09$	Reduction from Week 1 to Week 2
NSD (within intervention)	23	Wilcoxon signed-rank	$W = 16.0$	$p < .001$	$d_z = -1.28$	Reduction among participants with reliable coverage
Doomscrolling Scale (within intervention)	25	Wilcoxon signed-rank	$W = 7.0$	$p < .001$	$d_z = -1.25$	Reduction from Week 1 to Week 2

The supplementary outcomes show the same direction as the primary outcomes. Mean sessions per day and DSI both favored the intervention group, although they were not part of the four-outcome Holm family. The within-intervention comparisons also showed reductions from Week 1 to Week 2. These secondary results strengthen the descriptive pattern but remain exploratory.

4.4. SOP 3: Baseline Convergent Association Between Week 1 DSI and Self-Reported Doomscrolling Scores

Statement of the Problem 3 asked what baseline convergent association exists between the fixed-prior Week 1 Doomscroll Severity Index (DSI) and participants' self-reported doomscrolling scores among participants with at least three sentiment-reliable Week 1 sessions. The effective sample size was 48 because two intervention participants lacked sufficient reliable Week 1 sessions for DSI inclusion. Spearman's rho was treated as the primary statistic because the methodology selected a rank-based correlation when normality assumptions were not met.

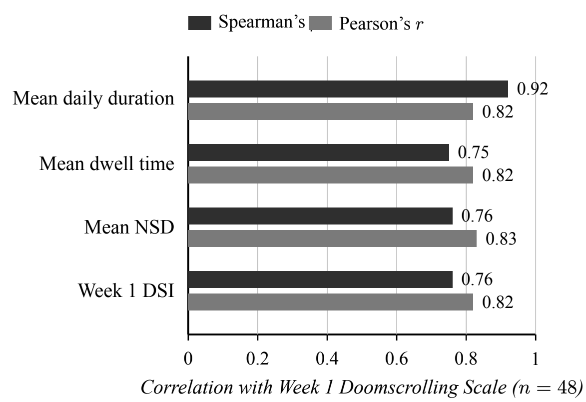
Table 15

SOP 3 Baseline Convergent Association with Week 1 Doomscrolling Scale

Predictor	n	Spearman's rho	Spearman p	Pearson's r	Interpretation
Week 1 mean daily duration	48	$\rho = 0.92$	$p < .001$	$r = 0.82$	Strongest rank association
Week 1 mean dwell time	48	$\rho = 0.75$	$p < .001$	$r = 0.82$	Strong positive association
Week 1 mean NSD	48	$\rho = 0.76$	$p < .001$	$r = 0.83$	Strong positive association
Week 1 DSI	48	$\rho = 0.76$	$p < .001$	$r = 0.82$	Strong convergent association

Figure 6

Baseline Convergent Associations Between Week 1 Behavioral Indicators and the Doomscrolling Scale



Every predictor achieved a correlation of at least 0.75 on both statistics, supporting convergent plausibility. Mean daily session duration produced the highest rank association ($\rho = 0.92$), while mean dwell time, mean NSD, and the composite DSI cluster together at $\rho \approx 0.75$-0.76. Pearson's r values are tightly grouped around $r = 0.82$-0.83, showing that the rank ordering, not just the linear relationship, drives the duration advantage.

The Week 1 DSI had a strong positive association with the Week 1 Doomscrolling Scale. This supports baseline convergent plausibility for the composite risk estimate. However, mean daily session duration had the highest rank correlation with self-report, so the data support convergence of the composite with self-reported doomscrolling rather than superiority over session duration alone.

4.5. SOP 4: User Evaluation Using the ISO/IEC 25010 Software Quality Model and the Technology Acceptance Model

Statement of the Problem 4 asked how users evaluated the system using the ISO/IEC 25010 software quality model and the Technology Acceptance Model (TAM). All six evaluated constructs met their pre-set favorable thresholds. The ISO/IEC 25010 Functional Suitability, Performance Efficiency, and Reliability subscales all exceeded the 3.50 favorable target. The SUS usability score exceeded the 70 target. TAM Perceived Usefulness and Perceived Ease of Use also exceeded 3.50.

Table 16

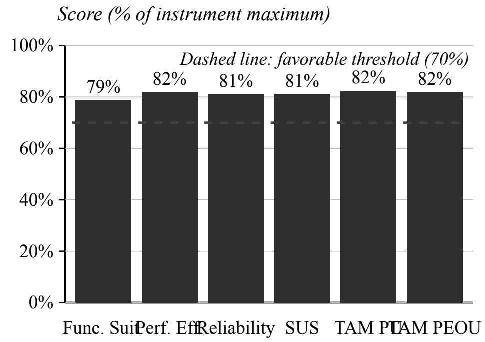
ISO/IEC 25010, SUS, TAM, and Reliability Summary

Construct	n	Items	Mean	SD	Alpha	Target
Functional Suitability	50	5	3.93	0.50	0.852	Met
Performance Efficiency	50	5	4.09	0.55	0.859	Met
Reliability	50	5	4.05	0.50	0.717	Met
SUS Usability	50	10	80.95	14.83	0.957	Met
TAM Perceived Usefulness	50	6	4.12	0.40	0.750	Met
TAM Perceived Ease of Use	50	6	4.09	0.53	0.855	Met

To place the six constructs on a shared axis, each score is expressed as a percentage of its instrument maximum (5-point Likert for ISO/IEC 25010 and TAM subscales; 0-100 for SUS). The 70% reference line represents the shared favorable threshold (3.50 out of 5 for Likert constructs, 70 out of 100 for SUS).

Figure 7

User Evaluation Construct Scores Against the Favorable Threshold



All six construct means cluster within a narrow band from 78.6% (Functional Suitability) to 82.4% (TAM Perceived Usefulness), and every bar sits comfortably above the 70% favorable line. The tight clustering suggests a uniformly positive user assessment across quality, usability, and acceptance dimensions rather than a strong preference for any single subscale.

The Cronbach's alpha values ranged from 0.717 to 0.957, meeting the study's descriptive benchmark of $\alpha \geq 0.70$ for every multi-item scale. These coefficients support internal consistency of the survey constructs in this sample, while still requiring caution because some subscales were researcher-developed or adapted to the specific system context.

Participant open-ended feedback provided practical context for the quantitative ratings. The most common coded theme was prompt timing, followed by onboarding/setup and score clarity. These themes show that users generally found the system useful and privacy-conscious, but also wanted clearer setup guidance, clearer score explanations, and more control over prompts.

Table 17

Participant Open-Ended Feedback Theme Summary

Theme	Count	Percent	Main Implication
Prompt timing	8	16%	Add snooze, quiet hours, or sensitivity controls
Onboarding and setup	6	12%	Improve Accessibility setup guidance

Theme	Count	Percent	Main Implication
Score clarity	6	12%	Explain the risk or activity score in simpler terms
Platform monitoring	5	10%	Show platform-specific monitoring status
Dashboard usefulness	5	10%	Retain and improve history and dashboard summaries
Breathing break	4	8%	Make the pause duration shorter or configurable
Export and study data	4	8%	Clarify which files are exported
Privacy and trust	4	8%	Keep local processing and explain stored data
Performance and battery	3	6%	Provide in-app battery and performance feedback
No major issue / general positive	5	10%	Some respondents reported no major difficulty

These qualitative findings support the survey results while also identifying improvement priorities. The strongest user-facing issues were not rejection of the system concept, but setup complexity, prompt tuning, score interpretation, and transparency around platform monitoring and exported data.

4.6. SOP 5: Subject Matter Expert Evaluation of Technical Design, Privacy Safeguards, Heuristic Logic, and Intervention Structure

Statement of the Problem 5 asked how subject matter experts evaluated the system's technical design, privacy safeguards, heuristic logic, and intervention structure. Two SMEs completed the evaluation: one software engineering or mobile application development expert and one digital well-being or behavioral psychology expert. Given the two-member panel, ratings are reported per expert against the study-defined favorable target of 4.00 rather than as an inferential panel statistic. The six rubric ratings cover technical soundness, input-range plausibility, rule-base coherence, architecture and privacy quality, intervention appropriateness, and overall ISO/IEC 25010 quality; the Overall column is the mean of these six ratings.

Table 18

Subject Matter Expert Rubric Results

Expert	Role	Tech.	Range	Rules	Privacy	Interv.	ISO	Overall
`SME-MOB-01`	Software engineering / mobile application development	5	5	5	5	5	5	5.00
`SME-PSY-01`	Digital well-being / behavioral psychology	4	5	4	4	5	4	4.33

Both SME ratings met the favorable target. The mobile/software expert rated all rubric areas as 5.00 and considered the hybrid VADER, fallback, and fuzzy inference approach appropriate for a non-clinical prototype, while recommending clearer operational definitions for aggregate-only handling and fallback triggers. The behavioral-psychology expert considered the Doomscrolling Feedback Loop Model an appropriate theoretical basis and supported the non-clinical self-monitoring framing. This expert also recommended softer terminology, clearer presentation of risk labels, and a shorter or configurable pause-and-reset duration.

The SME results support expert plausibility appraisal, not formal empirical calibration. The absence of a dedicated data science, machine learning, or fuzzy-logic reviewer remains a limitation. The expert comments identify concrete refinements: clearer fallback criteria, stronger privacy wording, less alarming labels, and more user control over interventions.

4.7. Analysis and Interpretation

The completed data show that the system met the study's software-acceptability thresholds, produced analyzable usage and sentiment logs across the target platforms, and showed favorable short-term differences in the intervention arm compared with the logging-only control arm. The results also show important boundaries. The pilot sample, two-week deployment window, purposive-convenience sampling, two-member SME panel, and sentiment-unreliable exclusions prevent broader claims about long-term efficacy, diagnostic validity, or population-level calibration.

Table 19

Summary of Findings by Statement of the Problem

SOP	Evidence	Main Finding	Interpretation and Limitation
1	Implementation, session logs, MVL concordance, data-quality summary, prompt logs, algorithm complexity	The privacy-preserving local architecture operated across the two-week deployment and produced 10,134 logged sessions with 87.0% sentiment-reliable coverage, with all core runtime components bounded to at most $O(n)$ per item and $O(1)$ per fuzzy inference.	Supports operational feasibility and bounded local risk estimation.
2	Final SOP 2 tests on Week 2 minus Week 1 change scores	All four primary outcomes favored intervention after Holm correction, with large effect sizes.	Supports observed short-term differences under study conditions, not long-term efficacy.
3	Week 1 DSI and Doomscrolling Scale correlation	DSI showed strong baseline convergence with self-report ($\rho = 0.76$), while duration had the highest rank association ($\rho = 0.92$).	Supports plausibility of DSI but not composite superiority or diagnostic validation.
4	ISO/IEC 25010, SUS, TAM, Cronbach alpha, user feedback	All six software-acceptability targets were met; alpha values were all at least 0.70.	Supports favorable user evaluation, with improvement needs in onboarding, prompt timing, score clarity, and monitoring transparency.
5	SME rubric and narrative feedback	Both SMEs met the favorable target, with overall means of 5.00 and 4.33.	Supports expert plausibility appraisal; limited by two experts and no dedicated fuzzy-systems reviewer.

Overall, the Chapter 4 evidence supports the completion of the thesis evaluation under the scope defined in Chapter 3. The system stands as a privacy-preserving, heuristic, non-clinical mobile tool that combines local behavioral logging, sentiment-aware risk estimation, and adaptive prompts, and the findings constitute promising pilot evidence within clear methodological limits.

Chapter 5

CONCLUSION

This chapter presents the summary of findings and conclusions based on the completed data analysis in Chapter 4. The conclusions are written within the study's actual scope: a pilot-scale software-engineering evaluation of a privacy-preserving Android system for heuristic doomscrolling-related risk estimation.

5.1. Summary of Findings

For the first research question, the study found that the implemented Android system could support privacy-preserving local risk estimation using observable behavioral and sentiment-related proxies. The final logs contained 10,134 sessions across TikTok, Facebook Reels, and Instagram Reels, with 8,817 sessions classified as sentiment-reliable and 1,317 sessions classified as sentiment-unreliable. The system implemented local processing, study-code-based export, text-first VADER-compatible sentiment analysis, Filipino/Taglish MVL support, a no-text VLM fallback, fuzzy inference, and conservative fallback handling. The MVL concordance data also showed exact agreement between the expert ratings and deployed runtime valences for all 57 runtime lexicon entries.

For the second research question, the intervention group showed larger Week 1-to-Week 2 reductions than the control group across the four primary outcomes. Session duration, video dwell time, NSD, and the self-reported Doomscrolling Scale all favored the intervention group after the final assumption-selected tests and Holm-Bonferroni correction. Supplementary results for sessions per day and DSI followed the same direction, and the within-intervention paired comparisons showed reductions from Week 1 to Week 2.

For the third research question, Week 1 DSI showed a strong positive baseline association with the Week 1 Doomscrolling Scale among the 48 eligible participants. The composite DSI had Spearman's $\rho = 0.76$, supporting convergent plausibility, while Week 1 mean daily session duration had the highest rank association with self-report at $\rho = 0.92$. The finding supports convergence between DSI and self-reported doomscrolling rather than superiority of the composite over its individual components.

For the fourth research question, the user evaluation met all pre-set software-acceptability targets. Functional Suitability, Performance Efficiency, Reliability, SUS Usability, TAM Perceived Usefulness, and TAM Perceived Ease of Use all reached the favorable thresholds. The SUS composite was 80.95, above the study target of 70. Cronbach's alpha values ranged from 0.717 to 0.957, meeting the $\alpha \geq 0.70$ internal-consistency benchmark. Open-ended feedback showed that users valued the dashboard, break reminders, privacy approach, and self-monitoring support, while also identifying needs around prompt timing, onboarding, score clarity, platform monitoring, breathing-break length, export transparency, and battery/performance feedback.

For the fifth research question, both subject matter experts rated the system favorably. The mobile/software expert gave an overall mean of 5.00, while the digital well-being/behavioral psychology expert gave an overall mean of 4.33. Their comments supported the non-clinical framing, privacy-conscious design, and general intervention structure, while recommending clearer fallback criteria, clearer stored-data wording, softer risk terminology, and more configurable intervention behavior.

5.2. Conclusions Based on Research Objectives and Questions

Based on the first research objective, the study concludes that the proposed system architecture was feasible for a bounded Android-based pilot deployment. The system was able to combine local monitoring, sentiment-aware processing, fallback handling, fuzzy inference, and adaptive prompts without requiring raw content retention in the research dataset. This supports the technical feasibility of the privacy-preserving design.

Based on the second research objective, the study concludes that the intervention condition was associated with favorable observed short-term differences in logged usage metrics and self-reported doomscrolling scores. These results support the promise of adaptive prompts as a short-term behavioral support mechanism under the study conditions; establishing long-term efficacy requires a longer deployment and a larger sample.

Based on the third research objective, the study concludes that the DSI has baseline convergent plausibility because it correlated strongly with the Doomscrolling Scale. Because session duration alone showed a stronger rank association in this sample, the DSI is best characterized as a transparent composite indicator that converges with self-report and still requires further validation and calibration.

Based on the fourth research objective, the study concludes that users evaluated the system favorably in terms of selected ISO/IEC 25010 characteristics, usability, and technology acceptance. The reliability coefficients also indicate acceptable to excellent internal consistency for the evaluation instruments in this sample. The qualitative feedback shows that the system was usable enough for pilot evaluation, but several parts of the user experience need refinement before broader deployment.

Based on the fifth research objective, the study concludes that the system received favorable expert plausibility appraisal from the available SME reviewers. The expert results support the appropriateness of the privacy framing, heuristic logic, and intervention structure for a non-clinical prototype. However, the two-expert panel and the absence of a dedicated fuzzy-systems or data-science reviewer limit the strength of the expert-validation claim.

Overall, the thesis concludes that the Heuristic Risk-State Estimation System is a defensible pilot prototype for privacy-preserving doomscrolling-related risk estimation and digital mindfulness prompting on Android short-form-video platforms. Its strongest contribution is the integration of local behavioral logging, text-first sentiment analysis, no-text fallback handling, fuzzy risk estimation, and user-facing prompts in one bounded software artifact. Its strongest limitation is that the evidence remains exploratory: the sample was purposive-convenience, the study lasted two weeks, some sentiment-dependent outcomes excluded unreliable sessions, and expert review was limited to two reviewers.

Chapter 6

RECOMMENDATIONS

This chapter presents recommendations derived from the findings, limitations, user feedback, expert comments, and implementation constraints discussed in Chapter 4 and Chapter 5. The recommendations focus on improving the current system, strengthening future versions, and guiding future researchers who may replicate or extend the study.

6.1. Recommendations for System Improvement

The system should include a guided onboarding flow for enabling the Android Accessibility Service and other required permissions. Onboarding and setup were among the most frequent participant feedback themes, and several respondents reported difficulty finding or understanding Android settings. A step-by-step setup guide with screen-specific instructions would reduce early friction and improve deployment consistency.

The application should explain the risk or activity score more clearly. User feedback showed recurring confusion about what the score meant and why some sessions received elevated labels. Future versions should show a simple component breakdown for session duration, video dwell time, and negative sentiment contribution. This would make the fuzzy score more transparent without exposing raw user content.

Prompt timing should be made more configurable. Since prompt timing was the most frequent open-ended feedback theme, the system should add options such as snooze, quiet hours, adjustable sensitivity, and clearer cooldown feedback. These changes would preserve the intervention function while giving users more control over interruptions.

The pause-and-reset or breathing-break feature should be shortened or made configurable. Both participant and expert feedback suggested that the breathing screen may feel too long in some contexts. A shorter default duration, or a user-selectable range, would better support autonomy while retaining the intended interruption effect.

The dashboard should include platform-monitoring status indicators for TikTok, Facebook Reels, and Instagram Reels. Some participants noticed inconsistent platform monitoring, especially around Instagram. A visible status indicator would help users understand whether each platform is actively being tracked, whether permissions are still enabled, and whether recent sessions were successfully logged.

The export feature should provide a preview or confirmation screen before sharing study data. Some participants were unsure which files were included in the export or whether the export had completed. The system should list exported file names, explain that raw text and screenshots are not included, and show a completion confirmation.

6.2. Recommendations for Future Development

Future versions should improve platform-specific extraction reliability. The system already logs reliability events, but future development should use those events to diagnose recurring extraction failures, high-OOV cases, and unresolved no-text cases by platform. This would help reduce sentiment-unreliable sessions and improve DSI coverage.

The VLM fallback should be further optimized and documented. Since unresolved VLM events contributed to reliability issues, future development should test prompt wording, model-response mapping, screenshot timing, and fallback thresholds. The goal should be to improve no-text handling while preserving local-only processing and avoiding persistent storage of screen frames.

The terminology used in the interface should be softened. Expert feedback recommended reducing alarming labels such as “Risk Score,” “Warning,” and “Critical.” Future versions may use terms such as “Activity Pattern Score,” “Elevated Use,” and “Extended Exposure,” as long as the thesis or documentation clearly maps these labels to the underlying analytic bands.

The system should retain the privacy-preserving local architecture. The favorable user and expert feedback around local processing indicates that privacy is a key strength of the system. Future enhancements should avoid unnecessary cloud processing and continue storing only aggregate metrics unless a new ethics review and consent procedure justifies a different design.

6.3. Recommendations for Future Researchers

Future researchers should conduct a longer deployment with a larger and more diverse sample. The present study used 50 purposively recruited participants over two weeks, which is appropriate for a pilot software evaluation but not enough for long-term efficacy or population-level claims. A longer study would help assess whether behavioral changes persist after novelty effects and initial prompting wear off.

Future researchers should include a dedicated data science, machine learning, or fuzzy-logic expert in the SME panel. The current two-expert panel provided useful software and behavioral feedback, but a dedicated fuzzy-systems reviewer would strengthen appraisal of membership functions, rule coherence, fallback thresholds, and defuzzification choices.

Future researchers should validate the DSI against additional external anchors. The present study found strong convergence with self-reported doomscrolling but also found that session duration had a higher rank association than the composite DSI. Future studies should compare DSI with full-scale doomscrolling instruments, ecological momentary reports, platform-independent digital well-being logs, or other validated behavioral measures.

Future researchers should evaluate whether the fuzzy composite adds explanatory value beyond simpler metrics. Because mean daily session duration showed the strongest baseline rank association with self-report, later studies should test whether DSI improves prediction, interpretability, or intervention timing after controlling for duration alone.

Future researchers should report effective sample sizes and sentiment coverage transparently. Sentiment-unreliable sessions affect NSD and DSI analysis, so future reports should continue to disclose retained sessions, excluded sessions, participant-level coverage, and platform-specific reliability patterns.

6.4. Recommendations for Future Implementation and Deployment

Future implementation should treat the system as a digital well-being support tool, not as a diagnostic or therapeutic application. The strongest defensible use case is self-monitoring and gentle interruption of prolonged short-form-video use. Any expansion toward mental-health claims would require a different clinical design, additional safeguards, and appropriate professional review.

Deployment should include clear consent and privacy explanations before enabling monitoring. Users should be told what the Accessibility Service reads, when transient screenshots may be used for no-text cases, what data are stored, what data are exported, and how they can stop monitoring or delete local data.

Deployment teams should prepare device-compatibility checks before field use. Android version differences, manufacturer background-process policies, and screenshot availability can affect monitoring stability. A short compatibility checklist should be completed before participants begin the baseline week.

Future deployments should include support materials for participants. A concise setup guide, troubleshooting guide, privacy FAQ, and export guide would reduce researcher support burden and help participants complete the study procedure consistently.

Finally, future implementation should keep evaluation claims proportional to the evidence collected. The current results support operational feasibility, favorable user evaluation, expert plausibility appraisal, and observed short-term differences under pilot conditions. Broader claims about long-term behavior change, diagnostic accuracy, or generalized effectiveness should be reserved for larger and longer studies with stronger validation designs.

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APPENDICES

APPENDIX A

Complete 27-Rule Fuzzy Inference Base

The following table presents the complete 27-rule inference base used by the three-input fuzzy engine (Video Dwell Time, Negative Sentiment Density, Session Duration). The 27 rows arise from exhaustive coverage of the $3 \times 3 \times 3$ input-state combinations. Their Safe/Warning/Critical outputs are fixed study priors intended for transparent heuristic inference rather than empirically calibrated probabilities or clinical thresholds. The assignment logic is conservative: Safe is reserved for low-baseline or isolated single-signal states, Warning is the default for mixed or moderate states, and Critical is reserved for combinations where the main risk axes converge under the fixed study priors.

Table 20

Full 27-Rule Fuzzy Inference Base

No.	Video Dwell Time	Negative Sentiment Density	Session Duration	Risk Level
1	Low	Low	Low	Safe
2	Low	Low	Medium	Safe
3	Low	Low	High	Warning
4	Low	Medium	Low	Safe
5	Low	Medium	Medium	Warning
6	Low	Medium	High	Warning
7	Low	High	Low	Warning
8	Low	High	Medium	Warning
9	Low	High	High	Critical
10	Medium	Low	Low	Safe
11	Medium	Low	Medium	Warning
12	Medium	Low	High	Warning
13	Medium	Medium	Low	Warning
14	Medium	Medium	Medium	Warning
15	Medium	Medium	High	Critical
16	Medium	High	Low	Warning

Table 19. Full 27-Rule Fuzzy Inference Base (continued)

No.	**Video Dwell Time**	**Negative Sentiment Density**	**Session Duration**	**Risk Level**
17	Medium	High	Medium	Critical
18	Medium	High	High	Critical
19	High	Low	Low	Safe
20	High	Low	Medium	Warning
21	High	Low	High	Warning
22	High	Medium	Low	Warning
23	High	Medium	Medium	Critical
24	High	Medium	High	Critical
25	High	High	Low	Warning
26	High	High	Medium	Critical
27	High	High	High	Critical

Interpretive note: the table follows a mostly non-decreasing assignment pattern in Session Duration and Negative Sentiment Density, while using Video Dwell Time as a contextual modifier rather than a standalone high-risk trigger. One deliberate exception is retained as a precautionary prototype rule for rapid negative-content chaining. Low dwell combined with high Negative Sentiment Density and high session duration is classified as **Critical** because rapid movement does not reduce concern when the session is both prolonged and clearly negative-dominant overall. By contrast, Rule 19 (High Dwell, Low NSD, Low Session Duration) stays **Safe** because sustained attention to one item, without prolonged use or negative-dominant exposure, is not enough by itself to imply doomscrolling. This keeps dwell time as a modifier, not the main driver.

Operationally, the 27 consequents follow a compact assignment grammar rather than 27 unrelated judgments: no rule is **Critical** when both Session Duration and Negative Sentiment Density are Low; duration alone does not produce **Critical** when NSD remains Low; High NSD reaches **Critical** only when reinforced by another non-low axis; High Dwell Time alone is insufficient for **Critical**; and **Warning** remains the default mixed class when the signals are concerning but not yet strongly convergent.

Read in groups, the table has three layers. Rules 1, 2, 4, 10, and 19 remain **Safe** because at least two axes stay low. Rules 3, 5-8, 11-14, 16, 20-22, and 25 remain **Warning** because they show mixed evidence without strong convergence between duration and negativity. Rules 9, 15, 17-18, 23-24, and 26-27 become **Critical** because prolonged use and negative exposure rise together, or because High NSD is reinforced by High Dwell Time in sessions that are no longer brief.

Within that Critical layer, the strongest challenge cases are still explainable. Rule 15 becomes **Critical** because a High-duration session with both other axes already at Medium no longer reflects time alone. Rule 23 becomes **Critical** because High Dwell Time, Medium NSD, and Medium Session Duration already show three non-low signals in the same direction. Rule 25 is retained as **Warning** after psychology expert review because high negative exposure and high dwell in a short session can be concerning, but the brief duration is not sufficient for the strongest intervention level.

APPENDIX B

Fallback Rules and Trigger Thresholds

Unless otherwise noted, the following thresholds are fixed study priors selected for reproducibility and conservative intervention behavior. They are design parameters, not empirically validated clinical or population cutoffs. Because the fallback engine retains two inputs with three linguistic levels each, exhaustive coverage requires $3^2 = 9$ rules.

In the implemented Android system, the two-input fallback rule base below is reserved for **Sentiment-Unreliable** sessions, specifically sessions where negativity cannot be resolved reliably after the available text or no-text path is attempted. This includes sessions whose extracted text is at least half unresolved vocabulary (study-defined unreliability trigger: session OOV ratio >= 50%) and sessions where the no-text VLM path cannot produce a stable item label because `AccessibilityService.takeScreenshot` is unavailable, screenshot capture fails, or VLM inference fails. The `>= 50%` value is not treated as a validated linguistic cutoff. It is a mathematically derived majority-representativeness screen: once unresolved tokens reach half of the valid tokens in a session, the recognized tokens no longer form the majority of the lexical content available to the text path. This is therefore used as a practical safeguard for noisy code-mixed text with spelling variation, lexical borrowing, and other textual variation rather than as a claimed language norm (Mohammed & Prasad, 2023; Perera & Caldera, 2024; Nazir et al., 2026; Khan et al., 2025). In those cases, the system degrades to a two-input fallback rule base (Video Dwell Time + Session Duration):

Table 21

Fallback 9-Rule Base (Sentiment-Unreliable Sessions)

No.	Video Dwell Time	Session Duration	Risk Level
1	Low	Low	Safe
2	Low	Medium	Safe
3	Low	High	Warning
4	Medium	Low	Safe
5	Medium	Medium	Warning
6	Medium	High	Warning
7	High	Low	Warning
8	High	Medium	Critical
9	High	High	Critical

Table 20. Fallback 9-Rule Base (Sentiment-Unreliable Sessions) (continued)

In Week 1 and in any participant without locked personalization, both fallback inputs use the global memberships reported in Appendix C. Once Week 2 personalization is locked, the fallback engine retains the same 9-rule consequents and RiskScore cutoffs but replaces only the Session Duration memberships with participant-specific Week 1 duration quantiles; Video Dwell Time remains fixed. This is because the fallback mode reuses the same live prompt engine's duration memberships while dropping only the sentiment input.

RiskScore cutoffs used across both full and fallback modes (equal-width reporting categories on the normalized 0-100 scale):

- Safe: 0.00 to 33.32
- Warning: 33.33 to 66.66
- Critical: 66.67 to 100

Prompt mapping (study-specified defaults derived from the reporting bands):

- Level 1: RiskScore 33.33 to 49.99
- Level 2: RiskScore 50.00 to 66.66
- Level 3 (Pause and Reset / Short Breathing Break): RiskScore 66.67 to 100 (disabled in fallback mode)

- The Warning band is split at its midpoint 50.00 so lower- and upper-Warning states receive different prompt intensities without introducing extra off-band cutoffs; Level 3 is reserved for the Critical band.
- Cooldown: fixed 15-minute prototype cooldown between prompts, intentionally reusing the same literature-bounded 15-minute live gate as the minimum re-eligibility interval so the system does not introduce a second unsupported minute constant; a repeat prompt therefore requires another prolonged interval of continued use. Recent JITAI work supports managing timing, receptivity, and dose (Teepe et al., 2021; Wang et al., 2023; Fiedler et al., 2024; Hsu et al., 2025; van Genugten et al., 2025)

APPENDIX C

Core Algorithmic Logic and Reproducibility Specifications

The Android implementation separates fixed analytic priors from Week 2 live-prompt personalization. Fixed global memberships are retained for week-level DSI computation across both study weeks so baseline and intervention summaries remain comparable. A separate live prompt engine is then allowed to replace only the Session Duration and NSD memberships after Week 1; Video Dwell Time remains fixed.

Fixed Analytic Priors (used for week-level DSI across both study weeks and as the Week 1 live default): Where

$$\mathrm{tri}(a, b, c)$$

is the standard triangular membership function, the study uses the following fixed memberships:

$$\begin{gathered}
\mathrm{Low}_{\mathrm{Dwell}} = \mathrm{tri}(0, 0, 5) \\
\mathrm{Medium}_{\mathrm{Dwell}} = \mathrm{tri}(4, 12, 20) \\
\mathrm{High}_{\mathrm{Dwell}} = \mathrm{tri}(15, 30, 30) \\
\mathrm{Low}_{\mathrm{NSD}} = \mathrm{tri}(0, 0, 33) \\
\mathrm{Medium}_{\mathrm{NSD}} = \mathrm{tri}(17, 50, 83) \\
\mathrm{High}_{\mathrm{NSD}} = \mathrm{tri}(67, 100, 100) \\
\mathrm{Low}_{\mathrm{Duration}} = \mathrm{tri}(0, 0, 10) \\
\mathrm{Medium}_{\mathrm{Duration}} = \mathrm{tri}(8, 15, 20) \\
\mathrm{High}_{\mathrm{Duration}} = \mathrm{tri}(15, 40, 40)
\end{gathered}$$

NSD values are clipped to [0, 100]. These fixed memberships govern week-level analytic DSI computation across both study weeks. In Week 1 they also act as the live default priors before any participant-specific personalization is available.

Where recent studies provide direct numeric anchors, only those anchors are treated as literature-bounded values: 5 s and 30 s for dwell time, 10 min, 15 min, and 40 min for session duration, and the normalized NSD/output partitions derived mathematically from the 0-100 scales. The remaining interior overlap points in the fixed priors are analytic smoothing values used to preserve overlap between adjacent triangular sets and prevent the engine from collapsing into hard step thresholds. They are therefore treated as transition parameters and included in the sensitivity analysis rather than defended as standalone validated behavioral cutoffs.

Week 2 Prompt Personalization (implemented in the live prompt engine only): For participant u, let the Week 1 sentiment-reliable-session values for personalized variable X (Session Duration or NSD) be $x_{u,1}, x_{u,2}, \ldots, x_{u,n_u}$. The prompt engine computes ordered quantiles:

$$Q_{25,u,X}, Q_{50,u,X}, Q_{75,u,X}, Q_{95,u,X}$$

The personalized Week 2 memberships are then:

$$\text{Low}_{X,\text{prompt},u} = \text{tri}(0, 0, Q_{50,u,X})$$

$$\text{Medium}_{X,\text{prompt},u} = \text{tri}(Q_{25,u,X}, Q_{50,u,X}, Q_{75,u,X})$$

$$\text{High}_{X,\text{prompt},u} = \text{tri}(Q_{50,u,X}, Q_{75,u,X}, Q_{95,u,X})$$

These quantiles are not treated as validated behavioral cutoffs. They are nonparametric order-statistic anchors chosen to map a three-term Low/Medium/High partition onto each participant's Week 1 distribution with the smallest transparent summary set: $\frac{\frac{Q_{25}}{Q_{50}}}{Q_{75}}$ summarize the interquartile core, while Q_{95} caps the High set without letting one raw maximum set the upper endpoint. Video Dwell Time remains fixed to the global priors above. If $n_u <$ 10 sentiment-reliable Week 1 sessions, the default priors are retained for that participant rather than locking personalized bounds. This 10-session minimum remains a numerical sufficiency safeguard rather than a validated population cutoff: below $n_u = 10$, the empirical Q_{95} is governed too heavily by the single largest observation or a nearly maximal order statistic, so the personalized High endpoint becomes too sensitive to one extreme session.

Recent personalized intervention-criteria work supports deriving live criteria from a first-week baseline, but recent JITAI reviews also note that empirical decision rules and points remain underdeveloped, which is why the implemented prompt engine uses participant-specific Week 2 criteria only when enough baseline data exist (Ikegaya et al., 2025; Hsu et al., 2025; van Genugten et al., 2025; Elmer et al., 2025). Because fallback prompting uses the same live prompt engine, personalized Session Duration bounds also carry into fallback mode when personalization is available; only the rule base and prompt cap differ.

Risk Score Category Cutoffs (0-100 scale):

- Safe: $0 \leq \text{RiskScore} < 33.33$
- Warning: $33.33 \leq \text{RiskScore} < 66.67$
- Critical: $66.67 \leq \text{RiskScore} \leq 100$

Output Centers Used in CoG Aggregation:

- Safe center: 16.67
- Warning center: 50
- Critical center: 83.33

These fixed centers are the band midpoints of the equal-width thirds above, so they are mathematically derived ordinal anchors rather than empirical risk probabilities. This removes arbitrary-looking output constants while preserving smooth interpolation across mixed-rule activations before the final category label is assigned.

Intervention Mapping and Cooldown:

- No intervention when RiskScore < 33.33
- Level 1 (Awareness Notification): $33.33 \leq \text{RiskScore} < 50$
- Level 2 (Pause Prompt): $50 \leq \text{RiskScore} < 66.67$
- Level 3 (Pause and Reset / Short Breathing Break): $66.67 \leq \text{RiskScore} \leq 100$

- The Warning band is split at its midpoint 50.00 so lower- and upper-Warning states receive different prompt intensities without introducing extra off-band cutoffs; Level 3 is reserved for the Critical band.
- Fixed cooldown: minimum 15-minute prototype interval between any two prompts, intentionally reusing the same literature-bounded 15-minute live gate as the minimum re-eligibility interval so the system does not introduce a second unsupported minute constant; a repeat prompt therefore requires another prolonged interval of continued use. Recent JITAI work supports managing timing, receptivity, and dose (Teepe et al., 2021; Wang et al., 2023; Fiedler et al., 2024; Hsu et al., 2025; van Genugten et al., 2025)

When the 2-input behavioral fallback mode is active (Sentiment-Unreliable), Level 3 is disabled by safety lock and the maximum permitted prompt is Level 2.

APPENDIX D

Filipino MVL and Neutral OOV-Reduction Inventory

The current Android implementation extends the VADER-compatible analyzer with the Filipino/Taglish Minimum Viable Lexicon (MVL) and neutral Filipino stop-word entries documented in the Filipino MVL review instrument. The Filipino MVL review was completed by one Filipino language expert and cross-checked against the deployed runtime lexicon in the workbook's `mvl_concordance` sheet. The review supports the deployed runtime inventory as a single-expert concordance check rather than as inter-rater validation. All 57 runtime entries (56 affective entries and one neutral candidate entry) matched the deployed valence values within the review window, and the neutral OOV-reduction terms were confirmed as sentiment-neutral. The neutral entries are assigned 0.0 valence only to reduce false OOV inflation from high-frequency Filipino function words; they do not add positive or negative sentiment to the compound score.

Table 22

Filipino/Taglish MVL Runtime Terms

Runtime group	Valence	Terms included in the runtime lexicon
Most negative	−4.0	`bobo`, `gago`, `kupal`, `poot`, `tanga`
Strongly negative	−3.0	`bwisit`, `dusa`, `galit`, `lungkot`, `masama`, `nakagagalit`, `nakakagalit`, `nakalulungkot`, `nakakalungkot`, `pangit`, `pighati`, `sakit`
Moderately negative	−2.0	`ayaw`, `hirap`, `inis`, `iyak`, `kasalanan`, `luha`, `mahirap`, `nakaiinis`, `nakakainis`, `nakatatakot`, `nakakatakot`, `selos`, `sisi`, `takot`, `talo`
Neutral candidate	0.0	`buhay`
Moderately positive	2.0	`aliw`, `ayos`, `gusto`, `ngiti`, `payapa`, `tawa`
Strongly positive	3.0	`bilib`, `galing`, `kilig`, `ligaya`, `lodi`, `mabait`, `maganda`, `mahal`, `masaya`, `paborito`, `pag-asa`, `petmalu`, `salamat`, `sigla`, `swerte`, `tiwala`, `tuwa`, `wagi`

The runtime list contains the Filipino/Taglish candidate terms used by the app and preserves common/formal spelling counterparts for four `naka-` roots. These counterparts are mapped to the same valence as their corresponding common form so that spelling variation does not unnecessarily increase the OOV ratio.

Table 23

Neutral Filipino OOV-Reduction Terms

Neutral term group	Valence	Terms counted as recognized but sentiment-neutral
Markers, linkers, and particles	0.0	ang, ng, sa, mga, na, ay, at, pa, si, ni, kay, lang, naman, ba, nga, din, rin, raw, daw, lamang
Demonstratives and locatives	0.0	ito, iyon, doon, dito, nito, niyan
Pronouns and possessives	0.0	nila, namin, ninyo, ako, ikaw, siya, kami, tayo, kayo, sila, ko, mo, niya, akin, iyo, kanila, kanya, ating, aming, inyong
Connectors and discourse terms	0.0	kapag, dahil, pero, kasi, upang, habang, bago, kaya, sana, kung, para, kapwa

APPENDIX E

Short-Form Doomscrolling Scale Administration Sheet

The study uses the short-form 4-item **Doomscrolling Scale**, corresponding to Items 1, 2, 10, and 12 of the original 15-item instrument developed by Sharma et al. (2022) and later confirmed in short-form use by Satici et al. (2023). To align self-report with each study week, the instrument is administered with a 7-day recall instruction stem. This modification narrows the recall window only; it does not create a separately validated weekly state instrument.

Participant Instructions: "In the past week, please indicate how strongly you agree with each statement."

Response Format:

- 1 = Strongly disagree
- 2 = Disagree
- 3 = Somewhat disagree
- 4 = Neither agree nor disagree
- 5 = Somewhat agree
- 6 = Agree
- 7 = Strongly agree

Items:

1. I feel an urge to seek bad news on social media, more and more often.
2. I lose track of time when I read bad news on social media.
3. I find myself continuously browsing negative news.

4. I feel like I am addicted to negative news.

Scoring Note: Item responses are summed or averaged to produce a single doomscrolling score, with higher values indicating greater self-reported doomscrolling during the recalled 7-day period. The 4-item short form contains no reverse-coded items.